A MODERN LOOK AT CLASSICAL GENERALIZATION BOUNDS

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Abstract. This thesis investigates the descriptive possibilities and limits of classical generalization bounds. We study this by comparing theoretical results with empirical evidence from experiments on binary decision trees under the zero-one loss. Our aim is to obtain a bound that qualitatively captures the behavior of the generalization gap as model complexity (maximum depth) increases and as aspects of the learning problem vary. The final result is a bound that accounts for the finite nature of our problem by counting hypotheses after observing the data. Moreover, it is sensitive to different noise levels in the data because it depends on the best risk attainable in the hypothesis class.

Si ringraziano il Prof. Savarè ed il Prof. Rosasco per essere delle guide fidate e costanti nel mio approccio al mondo della ricerca.

Un ringraziamento speciale a Pietro Zerbetto, senza cui il mio lavoro per questa tesi sarebbe stato infinitamente più difficile e meno divertente.

Infine, ringrazio tutti miei amici al MaLGa per avermi accolta e aver reso questa esperienza bella da morire.

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Research Context

This thesis was written during my stay at the Machine Learning Genoa Center (MaLGa) in Genova from July to November 2026, under the scientific supervision of Prof. Lorenzo Rosasco. Building on my interest in the mathematical foundations of machine learning, Prof. Rosasco proposed studying classical generalization bounds and empirically assessing how well they capture the qualitative behavior of the generalization gap. His initial proposal and scientific guidance shaped the present work.

1. Introduction

In machine learning, there exists a large number of algorithms, notably neural networks trained with gradient descent, which achieve great performance at prediction tasks on unseen data. The process of applying a model that has been learned on some available data to new, unseen data is called *generalization*. A central question in learning theory is what drives the generalization ability of a learning algorithm [23, 19]. Several works have addressed this question, suggesting that classical theory does not fully explain generalization. In particular, this means that it can fail to justify the behavior of the test error as the complexity of the hypothesis class increases [6, 23, 24, 18].

This thesis contributes to this discussion by testing classical theoretical results against empirical evidence. Specifically, we study generalization bounds. A generalization bound is a bound on the *generalization gap*, i.e. the difference between the true and the empirical risk of a model selected on some available data D,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \tag{1}$$

where the expectation is taken over the disorder of the data. Then, we inquire into the empirical validity of these bounds as functions capturing the *qualitative behavior* of the empirical generalization gap as model complexity increases. Unlike the traditional use of these results, we are not concerned with finding a quantitatively exact bound on the generalization gap, but one that reproduces its shape. To do so, we run experiments in the setting of a finite hypothesis class of binary decision trees under the zero-one loss. We look for a function that increasingly accounts for the structural aspects of our learning problem. The final result of this thesis is a bound on the generalization gap of decision trees that captures its saturating behavior and that is sensitive to the noise level in the data, summarized in the following theorem.

Theorem 1.1. *Let $R^*_{\mathscr{H}} = \min_{h \in \mathscr{H}} R(h)$ be the best risk attainable by hypotheses. Then*

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \leq 4\sqrt{\frac{(R^*_{\mathscr{H}})\,(\log(4\Pi_{2n}(\mathscr{H})) + 1)}{n}} + 20\,\frac{\log(4\Pi_{2n}(\mathscr{H})) + 1}{n}$$

with

$$\Pi_{2n}(\mathscr{H}) \leq \min\left\{|\mathscr{H}|, 2^{2n}\right\}.$$

Hence, the main contribution of this work is to provide an empirical analysis of the limits and possibilities of classical bounds as descriptive tools. Hopefully, we also develop a framework and approach that can be extended to the study of more advanced generalization bounds, as well as other learning algorithms, like gradient-based ones.

1.1. **Related Work.** This section presents the state of the art and how this thesis contributes to it.

On the empirical evaluation of generalization bounds. Existing empirical studies of generalization bounds for decision trees mainly compare their numerical tightness [12, 20] or use them for pruning and model selection [13, 15, 22]. Our interest is different: we consider hypotheses sets $\mathscr{H}_d$ fixed in advance and compare the shape of analytical bounds with the empirical generalization gap as tree depth increases. Other works refine bounds through dependence on the training sample or an effective number of leaves [16, 9, 5], without empirically evaluating their descriptive power. Outside decision trees, [11] investigate how complexity measures relate to generalization in neural networks. We share this interest in comparing theory with observed generalization, focusing on what classical bounds can explain for decision trees.

Theoretical surveys of generalization bounds. The survey [7] has been a fundamental guide for the theoretical side of this thesis. Its emphasis, also shared by statistical analyses of CART [14], is on how risk bounds behave as the sample size n grows. We, instead, keep n fixed and study bounds as functions of the complexity of the hypotheses set. This changes the role of the theoretical tools: our Theorem 5.2, analogous to Theorem 5.1 of [7], allows us to account for noise, whereas the survey mainly uses it to improve the dependence on n when the training error is small.

In light of the related literature, the contribution of this work is to investigate the limits and possibilities of classical generalization bounds as descriptive tools. We use a modern approach entailing empirical tests, compared to the traditional

literature on generalization bounds; moreover, we take a different viewpoint that considers bounds as functions capturing the qualitative behavior of the curve for descriptive purposes, instead of risk guarantees or model selection.

1.2. **Setting and Approach.** Given a sample space Ω, let inputs and labels be generated by $X : \Omega \to \mathcal{X} = \{\bar{x}_1, \ldots, \bar{x}_N\}$ and $Y : \Omega \to \mathcal{Y} = \{-1, 1\}$. Let p be the joint probability distribution of (X, Y) on $\mathcal{X} \times \mathcal{Y}$, with

$$p_{j,y} = \Pr[X = \bar{x}_j, Y = y], \qquad p_{j,y} \geq 0, \qquad \sum_{j=1}^{N} \sum_{y \in \mathcal{Y}} p_{j,y} = 1.$$

Let $D = ((X_i, Y_i))_{i=1}^n$ be a dataset of n i.i.d. copies of (X, Y) drawn from p. We use the term *dataset* to refer to the *random variable* generating the data throughout this work. We denote by $S = (x_i, y_i)_{i=1}^n$ a realization of it. Given this data, we want to learn a classifier $h : \mathcal{X} \to \mathcal{Y}$ that given a new input $x \in \mathcal{X}$ classifies it correctly. Let $\mathscr{H} \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$ be a nonempty, finite set of functions where we look for such a classifier. Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$, let

$$R(h) \coloneqq \mathbb{E}(\ell(h(X), Y)) = \sum_{\bar{x} \in \mathcal{X}} \sum_{\bar{y} \in \{-1,1\}} \ell(h(\bar{x}), \bar{y}) p(\bar{x}, \bar{y}).$$

be the true risk (or test error), and

$$\hat{R}_D(h) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \ell(h(X_i), Y_i).$$

the empirical risk (or training error), for a fixed n. Because we do not have access to p, we select the *empirical risk minimizer* (ERM),

$$\hat{h}_D \in \arg\min_{h \in \mathscr{H}} \hat{R}_D(h) \tag{2}$$

which, importantly, is a random variable of the data D. Note that at least one such $\hat{h}_D$ exists, since $\mathscr{H}$ is finite and nonempty.

We study bounds on the expected difference between the test and train error of the ERM,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \tag{3}$$

that is called *generalization gap*. We state our first results in a general setting, for arbitrary hypothesis classes and loss functions. As our study develops, we use

results that are specific to the zero-one loss, i.e.

$$\ell(h(x), y) = \mathbf{1}(h(x) \neq y) = \frac{1 - h(x)y}{2}, \tag{4}$$

that is also used for the experiments. On the experimental side, in fact, we consider the algorithm of binary decision trees with decision rules at the nodes of type $x \geq t$ for $t \in \mathbb{R}$, trained with the zero-one loss. So, for a fixed depth d, our hypothesis class $\mathscr{H}_d$ is the class of functions representable by binary decision trees of depth at most d. This is the class of piecewise constant functions $h : \mathcal{X} \to \{1, -1\}$ with at most 2^d contiguous blocks, i.e.

$$\mathscr{H}_d \coloneqq \left\{ h : \mathcal{X} \to \{1, -1\} : \sum_{j=1}^{N-1} \mathbf{1}_{h(\bar{x}_j) \neq h(\bar{x}_{j+1})} \leq 2^d - 1 \right\} \qquad \text{for } d > 0. \tag{5}$$

It is useful to define the following quantities. For a fixed n and variable d, let

$$\mathcal{T}(d) \coloneqq \mathbb{E}[R(\hat{h}_{D,d})], \qquad \mathcal{L}(d) \coloneqq \mathbb{E}[\hat{R}_D(\hat{h}_{D,d})], \qquad \mathcal{G}(d) \coloneqq \mathcal{T}(d) - \mathcal{L}(d)$$

be the true risk, the (expected) empirical risk, and the generalization gap of decision trees with depth at most d respectively. In this notation, we look for functions $\mathcal{B}(d)$ such that

$$\mathcal{G}(d) \lesssim \mathcal{B}(d),$$

from which

$$\mathcal{T}(d) \lesssim \mathcal{B}(d) + \mathcal{L}(d).$$

A bound on the gap is a bound on the true risk when $\mathcal{L}(d)$ is 0 or negligible. For visualization purposes, we obtain the curves $\mathcal{B}(d)$ by heuristically rescaling the bounds stated in the theorems to make them comparable in magnitude to the empirical curves. This reflects our focus on the qualitative dependence of the bounds on d, rather than on their numerical values.

The rest of this work is organized as follows. Section 2 provides some preliminary results used throughout. In Sections 3 to 5, we present theoretical results, describe the experimental design and plot empirical curves and qualitative bounds, and then comment on the results. Finally, in Section 6 we conclude and discuss limitations and potential developments of our analysis. Appendices A and B contain some preliminary experiments and additional details on the experimental design, respectively.

The code is available at https://github.com/matildedol/tesi.

2. Preliminary Results

In this section, we state some preliminary results that are central to the results developed in the following sections.

The following is a fundamental result on the moment generating function of a random variable.

Theorem 2.1 (Hoeffding's lemma)**.** *[*[8]*, Lemma 2.2] Let Z be a real-valued random variable with $\mathbb{E}[Z] = 0$ and $|Z| \leq a$ almost surely for some $a \in \mathbb{R}$. Then, for every $t \in \mathbb{R}$,*

$$\mathbb{E}\left[e^{tZ}\right] \leq e^{t^2 a^2/2}.$$

The reference states it for $Z \in [a', b']$ with bound $e^{t^2(b'-a')^2/8}$; here $[a', b'] = [-a, a]$. See also [10].

The following theorems are established concentration inequalities, i.e. upper bounds on the deviation of a random variable from its expected value. While Hoeffding's inequality mainly uses the bounded interval over which the random variable ranges, Bernstein's and Chebyshev's are more refined results using its second moment.

Theorem 2.2 (Hoeffding's inequality)**.** *[*[10]*, Theorem 2], [*[8]*, Theorem 2.8] Let $Z_1, \ldots, Z_n$ be independent real-valued random variables with $Z_i \in [a_i, b_i]$ almost surely for some $a_i, b_i \in \mathbb{R}$, and let $\mu = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i]$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} Z_i - \mu \geq \varepsilon\right] \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

Theorem 2.3 (Chebyshev's inequality)**.** *[*[8]*, Section 2.1] Let Z be a real-valued random variable with finite variance. Then, for every $\varepsilon > 0$,*

$$\Pr\left[|Z - \mathbb{E}[Z]| \geq \varepsilon\right] \leq \frac{\mathrm{Var}(Z)}{\varepsilon^2}.$$

Theorem 2.4 (Bernstein's inequality)**.** *[*[8]*, Theorem 2.10] Let $Z_1, \ldots, Z_n$ be independent real-valued random variables with $\mathbb{E}[Z_i] = 0$ and $Z_i \leq M$ almost surely for some $M > 0$, and let $v = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Var}(Z_i)$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} Z_i \geq \varepsilon\right] \leq \exp\left(-\frac{n\varepsilon^2}{2(v + M\varepsilon/3)}\right).$$

The reference bounds $\Pr[\sum_i Z_i \geq t]$ in terms of $\sum_i \mathbb{E}[Z_i^2]$, which equals nv since $\mathbb{E}[Z_i] = 0$; the statement above follows by taking $t = n\varepsilon$.

The following proposition is the main tool used in Theorem 3.1 that provides the first generalization gap considered in this study.

Proposition 2.5 (Union bound)**.** *[*[17]*, Appendix C] For any finite or countable collection of events $A_1, A_2, \ldots$,*

$$\Pr\left[\bigcup_i A_i\right] \leq \sum_i \Pr[A_i].$$

Tail integration is a crucial tool which allows us to go from results on the probability that a random variable is large, like the ones above, to results on its *expected* value.

Proposition 2.6 (Tail integration)**.** *[*[21]*, Lemma 1.2.1] For any non-negative random variable Z,*

$$\mathbb{E}[Z] = \int_0^\infty \Pr[Z \geq \varepsilon] \ \mathrm{d}\varepsilon.$$

The reference uses $\Pr[Z > \varepsilon]$; the two integrands differ on at most countably many ε, hence the integrals coincide.

Finally, the Gaussian tail bound is used in the proof of Theorem 3.1.

Proposition 2.7 (Gaussian tail bound)**.** *[*[21]*, Proposition 2.1.2] For every $c > 0$ and $x_0 > 0$,*

$$\int_{x_0}^\infty e^{-cx^2} \ \mathrm{d}x \leq \frac{1}{2cx_0} e^{-cx_0^2}.$$

The reference states $\int_t^\infty e^{-u^2/2} \ \mathrm{d}u \leq \frac{1}{t} e^{-t^2/2}$ for $t > 0$; the statement above follows by the change of variables $u = \sqrt{2c}\, x$, $t = \sqrt{2c}\, x_0$.

Assumptions on the loss function

Consider the following assumption on the loss function ℓ.

Assumption 2.8 (Boundedness of loss function)**.** *Given a set $A \subseteq \mathbb{R}$ and $\ell : A \times A \to \mathbb{R}_+$, $\exists M \in \mathbb{R}, M > 0$ s.t.*

$$\ell(a, a') \leq M \qquad \forall a, a' \in A$$

A bounded loss function is crucial for obtaining all the upper bounds on the gap discussed in this work.

Fact 2.9. *The zero-one loss satisfies Assumption 2.8 with $M = 1$.*

Fact 2.10. *If the label space $\mathcal{Y}$ is finite, then any loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$ is bounded.*

The previous facts show that Assumption 2.8 is always satisfied in our setting.

Nonetheless, we now give a sufficient condition that allows us to circumvent boundedness. This is useful for generalizing the results obtained in this work to other learning algorithms with real-valued predictors.

Assumption 2.11 (Lipschitzness of loss function and bound at zero)**.** *Given two sets $A, B \subseteq \mathbb{R}$ and $\ell : A \times B \to \mathbb{R}_+$,*

(i) $\exists C > 0$ *s.t.*

$$|\ell(a, b) - \ell(a', b')| \leq C\,(|a - a'| + |b - b'|) \qquad \forall a, a' \in A,\ b, b' \in B$$

(ii) $\exists C' > 0$ *s.t.*

$$|\ell(0, 0)| \leq C'$$

Proposition 2.12. *Let $A, B \subseteq [-K, K]$ for some constant $K > 0$, with $0 \in A$, $0 \in B$. Then Assumption 2.11 $\Rightarrow$ Assumption 2.8.*

Proof. Let $a \in A$ and $b \in B$. By the triangle inequality and the two points of Assumption 2.11, applied with $(a', b') = (0, 0)$,

$$|\ell(a, b)| \leq |\ell(0, 0)| + |\ell(a, b) - \ell(0, 0)| \leq C' + C\,(|a| + |b|) \leq C' + 2CK.$$

Since $\ell \geq 0$, Assumption 2.8 holds with $M = C' + 2CK > 0$. □

Therefore, if the inputs are bounded, Lipschitzness is sufficient to obtain boundedness.

On top of this, let $T : \mathbb{R} \to [-1, 1]$,

$$T(a) = \begin{cases} -1 & a \leq -1 \\ a & -1 < a < 1, \\ 1 & a \geq 1 \end{cases} \qquad a \in A \tag{6}$$

be a *truncation map*. For any hypothesis class $\mathscr{H} = \{h : \mathcal{X} \to \mathcal{Y}\}$, we can always define $\mathscr{H}_T = \{h_T \coloneqq T \circ h : \mathcal{X} \to [-1, 1] | h \in \mathcal{H}\}$ and take $A = \{(h(x), y) : h \in \mathscr{H}_T, x \in \mathcal{X}, y \in \mathcal{Y} = \{1, -1\}\}$ which satisfies the assumption of bounded inputs required by Proposition 2.12.

Therefore, Proposition 2.12 and the truncation technique determine that Lipschitzness of the loss function immediately implies boundedness. The former is easier to verify, as Example 2.13 shows, and does not depend on the hypothesis class.

Example 2.13 (Logistic loss function)**.** *Let $\ell(a, b) = \log(1 + e^{-ab})$, $a \in A$, $b \in B$, with $A, B \subseteq [-K, K]$ for some $K > 0$, be the logistic loss function. Its partial derivatives are*

$$\frac{\partial \ell}{\partial a}(a, b) = -\frac{b}{1 + e^{ab}}, \qquad \frac{\partial \ell}{\partial b}(a, b) = -\frac{a}{1 + e^{ab}},$$

hence, since $\frac{1}{1+e^{ab}} < 1$,

$$\left|\frac{\partial \ell}{\partial a}(a, b)\right| \leq |b| \leq K, \qquad \left|\frac{\partial \ell}{\partial b}(a, b)\right| \leq |a| \leq K \qquad \forall a \in A,\ b \in B.$$

By the mean value theorem applied to each argument separately,

$$|\ell(a, b) - \ell(a', b')| \leq |\ell(a, b) - \ell(a', b)| + |\ell(a', b) - \ell(a', b')| \leq K \left(|a - a'| + |b - b'|\right),$$

so $C = K$. With the truncation map and $\mathcal{Y} = \{1, -1\}$ we have $K = 1$, hence $C = 1$. Also,

$$\ell(0, 0) = \log 2 = C'.$$

3. A First Approach

In this section, we present our first bound on the generalization gap and we compare it with the empirical curve in the setting of binary decision trees.

Theorem 3.1. *Let $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$ be a loss function bounded by $M \in \mathbb{R}_+$. Then*

$$0 \leq \mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq 2M\sqrt{\frac{\log|\mathscr{H}|}{2n}}$$

Source. Adapted from [17], Theorem 2.13: the original is a high-probability bound for a finite class; here we state it in expectation, using tail integration, and we add the lower bound, which is specific to the ERM.

Proof of Theorem 3.1. We prove the lower bound first. Since $\hat{h}_D \in \arg\min_{h\in\mathscr{H}} \hat{R}_D(h)$, we have

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h) \qquad \forall h \in \mathscr{H},$$

and taking expectations

$$\mathbb{E}[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}[\hat{R}_D(h)] \qquad \forall h \in \mathscr{H}.$$

For a fixed $h \in \mathscr{H}$, which does not depend on D, the empirical risk is an unbiased estimate of the true risk, by linearity and because the (X_i, Y_i) are distributed as (X, Y),

$$\mathbb{E}[\hat{R}_D(h)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\ell(h(X_i), Y_i)\right] = R(h).$$

Hence

$$\mathbb{E}[\hat{R}_D(\hat{h}_D)] \leq R(h) \qquad \forall h \in \mathscr{H}.$$

Now take S a realization of D and consider in particular $h = \hat{h}_S \in \mathscr{H}$,

$$\mathbb{E}[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_S).$$

Since this holds for every realization S of D, we have $\mathbb{E}[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_D)$ almost surely and, taking expectations,

$$\mathbb{E}[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}[R(\hat{h}_D)].$$

Next, we prove the upper bound. Fix $h \in \mathscr{H}$ and let $Z_i \coloneqq -\ell(h(X_i), Y_i)$, $i = 1, \dots, n$, so that

$$\frac{1}{n}\sum_{i=1}^{n} Z_i - \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i] = (R - \hat{R}_D)(h).$$

We want to apply Hoeffding's inequality (Theorem 2.2). The Z_i are independent, since the (X_i, Y_i) are, and $Z_i \in [-M, 0]$ by non-negativity of the loss function and Proposition 2.12. Hence, with $a_i = -M$ and $b_i = 0$, we obtain

$$\Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \leq e^{-2n\varepsilon^2/M^2}. \tag{7}$$

Now, the ERM $\hat{h}_D$ is a random variable dependent on the data D. To bound its deviation, we require the concentration event to hold simultaneously for all functions $h \in \mathscr{H}$. Therefore, by the union bound (Proposition 2.5) and (7),

$$\begin{aligned}
\Pr\left[(R - \hat{R}_D)(\hat{h}_D) \geq \varepsilon\right] &\leq \Pr\left[\bigcup_{h \in \mathscr{H}} \left\{(R - \hat{R}_D)(h) \geq \varepsilon\right\}\right] \\
&\leq \sum_{h \in \mathscr{H}} \Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \\
&\leq |\mathscr{H}| e^{-2n\varepsilon^2/M^2}.
\end{aligned} \tag{8}$$

To obtain a bound in expected value from (8), we tail integrate. Let $X \coloneqq (R - \hat{R}_D)(\hat{h}_D)$ and $X^+ \coloneqq \max\{X, 0\}$, which is non-negative, so that by Proposition 2.6

$$\mathbb{E}[X] \leq \mathbb{E}[X^+] = \int_0^\infty \Pr[X^+ \geq \varepsilon] \ \mathrm{d}\varepsilon = \int_0^\infty \Pr[X \geq \varepsilon] \ \mathrm{d}\varepsilon.$$

Let $\alpha \coloneqq |\mathscr{H}|$ and $c \coloneqq 2n/M^2$, so that (8) reads $\Pr[X \geq \varepsilon] \leq \alpha e^{-c\varepsilon^2}$. This bound is good for large values of ε, but for small ε it exceeds 1, hence we use instead

$$\Pr[X \geq \varepsilon] \leq \min\{1, \alpha e^{-c\varepsilon^2}\}.$$

We split the integral at the threshold ε_0 where $\alpha e^{-c\varepsilon_0^2} = 1$, i.e. $\varepsilon_0 = \sqrt{\log \alpha / c}$. Using the Gaussian tail bound (Proposition 2.7) for the second integral, we obtain

$$\mathbb{E}[X] \leq \int_0^{\varepsilon_0} 1 \ \mathrm{d}\varepsilon + \int_{\varepsilon_0}^\infty \alpha e^{-c\varepsilon^2} \ \mathrm{d}\varepsilon \leq \varepsilon_0 + \frac{\alpha}{2c\varepsilon_0} e^{-c\varepsilon_0^2} = \sqrt{\frac{\log \alpha}{c}} + \frac{1}{2\sqrt{c \log \alpha}},$$

where in the last equality we used $\alpha e^{-c\varepsilon_0^2} = 1$. Finally, assuming $\alpha \geq e^{1/2}$, which holds as soon as $|\mathscr{H}| \geq 2$, we have $\frac{1}{2\sqrt{c \log \alpha}} \leq \sqrt{\frac{\log \alpha}{c}}$, and hence

$$\mathbb{E}[X] \leq 2\sqrt{\frac{\log \alpha}{c}} = 2M\sqrt{\frac{\log |\mathscr{H}|}{2n}}.$$

It remains to consider the case $|\mathscr{H}| = 1$, where the bound reads 0 and the argument above does not apply ($\log \alpha = 0$ makes $\varepsilon_0 = 0$ and the last step fails). If $\mathscr{H} = \{h\}$, then $\hat{h}_D = h$ for every D, which does not depend on the data, so by the unbiasedness of the empirical risk for a fixed hypothesis shown above

$$\mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] = R(h) - \mathbb{E}[\hat{R}_D(h)] = 0,$$

and both inequalities hold with equality. □

This result formally demonstrates that the empirical risk of the ERM systematically underestimates its true risk, with the generalization gap scaling as $\mathcal{O}\left(\sqrt{\frac{\log |\mathscr{H}|}{n}}\right)$. The logarithmic dependence on $|\mathscr{H}|$ arises by first establishing the exponential concentration of $\hat{R}_D(h)$ for a fixed hypothesis $h \in \mathscr{H}$ with Hoeffding's inequality, then taking a union bound over the class of functions $\mathscr{H}$, and tail integrating.

Note that while the upper bound is valid for any hypothesis $h \in \mathscr{H}$, the lower bound is specific to the ERM. We next present the experimental design to evaluate the upper bound for binary decision trees and compare it with the empirical test error curve on a qualitative level.

Experimental design

Data-generating process. We create the support $\mathcal{X}$ as a discretization of the open interval $(0, 10)$ into $N = 10^9$ evenly spaced points. We set N large enough so that it resembles the continuous case. Then, we sample from $\mathcal{X}$ with the uniform measure $p_{\mathcal{X}}(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p(\bar{x}, \bar{y}) = \frac{1}{N}$. The law to generate the labels is

$$y \leftarrow \operatorname{sign}(f^*(x) + \sigma\varepsilon)$$

where $\operatorname{sign}(x) = 1$ if $x \geq 0$ and -1 otherwise. $f^* : \mathcal{X} \to \mathbb{R}$ is the data-generating function, $\varepsilon \sim \mathcal{N}(0, 1)$ is a Gaussian noise, and $\sigma > 0$ controls the noise level. First, we set f^* to a centered parabola, so that roughly half of the labels are 1 and half -1 in the interval $(0, 10)$. Since an algorithm could learn such a function with few datapoints (see preliminary experiments in Appendix A.2), we switch to a sinusoid with varying frequency. Namely, we use

$$f^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\, x\right) \tag{9}$$

where m is the number of times f^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of 1s and -1s. We set as default value $m = 10$.

We test a Bernoullian noise instead of a Gaussian one, concluding that there is no interesting difference in the performance of the algorithm as long as the portion of misclassified inputs due to the noise is the same (see preliminary experiments in Appendix A.1). Hence, we fix the noise level to 20% and find that for $m = 10$, $\sigma \approx 3.5$ achieves it on average.

All in all, our labels are generated as

$$y \leftarrow \operatorname{sign}\left(5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right) + \sigma\varepsilon\right)$$

with $\sigma \approx 3.5$ and $m = 10$. All the hyperparameters of the experiments are summarized in Table 1 in Appendix B.1.

Experiment. We generate a test set with size $n_{\text{test}} = 200,000$ sampling from $\mathcal{X}$ with $p_{\mathcal{X}}$ and with replacement, and generating the labels as above. We select a range of depths that allows us to observe the asymptotic behavior of the empirical curves, namely up to 30. Then, for each depth $d = 1, \ldots, 30$, for $T = 100$ iterations, we

(1) generate a train set ($n = 500$);

(2) fit a decision tree with depth at most d (DecisionTreeClassifier algorithm from the scikit-learn library);

(3) compute the train error, the test error, and the difference between the two (the gap).

Then, we average the resulting quantities over the T samples, obtaining empirical estimates of the expected values of the train error, test error, and gap over the disorder of the data.

Learning algorithm. To fit decision trees, we use the DecisionTreeClassifier algorithm from the scikit-learn library. It uses the same decision rule at the nodes that we assumed in the theoretical part of this thesis, namely $x \geq t$ for arbitrary $t \in (0, 10)$. However, it is a greedy algorithm, which does not always select the ERM (2). See Section 6 for a discussion of this difference.

Cardinality of the hypothesis class. To plot the theoretical bound, instead, we need to express the cardinality $|\mathscr{H}_d|$ in terms of the maximum depth d.

Fact 3.2 (Upper bound on $|\mathscr{H}_d|$)**.** *Since* $\mathscr{H}_d \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$, $|\mathcal{X}| = N$ *and* $|\mathcal{Y}| = 2$, *we have*

$$|\mathscr{H}_d| \leq 2^N \qquad \forall d > 0.$$

Recall the definition of the hypothesis class (5). A function with exactly r blocks is determined by the choice of the label in the first block (2 options) and by the position of the remaining $r-1$ thresholds, which are chosen among the $N-1$ intervals $(\bar{x}_j, \bar{x}_{j+1})$ for $j = 1, \ldots, N-1$: there are $\binom{N-1}{r-1}$ possible such functions. Summing over $r = 1, \ldots, 2^d$ and imposing $r \leq N$, we have

$$|\mathscr{H}_d| = 2 \sum_{j=0}^{\min\{2^d-1, N-1\}} \binom{N-1}{j}.$$

Note that for $2^d > N$,

$$|\mathscr{H}_d| = 2 \sum_{j=0}^{N-1} \binom{N-1}{j} = 2^N$$

attaining the upper bound given in Fact 3.2. For $2^d \ll N$, instead, Stirling's approximation gives

$$\log |\mathscr{H}_d| \approx (2^d - 1) \log \frac{eN}{2^d}.$$

Thus,

$$\log |\mathscr{H}_d| \approx \min \left((2^d - 1) \log \frac{eN}{2^d}, N \right). \tag{10}$$

Bound. We refer to the final bound as *Union bound*. This is a scaled version of the bound provided by Theorem 3.1,

$$\mathcal{B}_{Union}(d) = C_{Union} \cdot 2 \sqrt{\frac{\log |\mathscr{H}_d|}{2n}} \tag{11}$$

with $C_{Union} = 0.0006$ so that $0 \leq \mathcal{B}_{Union}(d) \leq 1$ for $1 \leq d \leq 30$, the depth range we consider. See Appendix B for details on how this constant is fitted.

Empirical analysis

First, in Figure 1 we plot the train and test error of binary decision trees as depth increases.

We observe that both the test and train error saturate, and that the train error goes approximately to 0. Indeed, in the following we also inspect the case $\mathcal{T}(d) \lesssim \mathcal{B}(d) + \mathcal{L}(d)$.

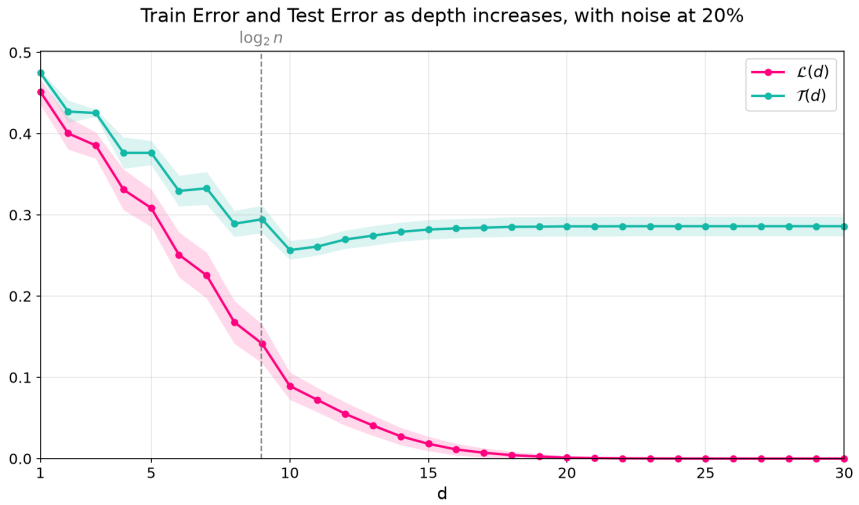


FIGURE 1

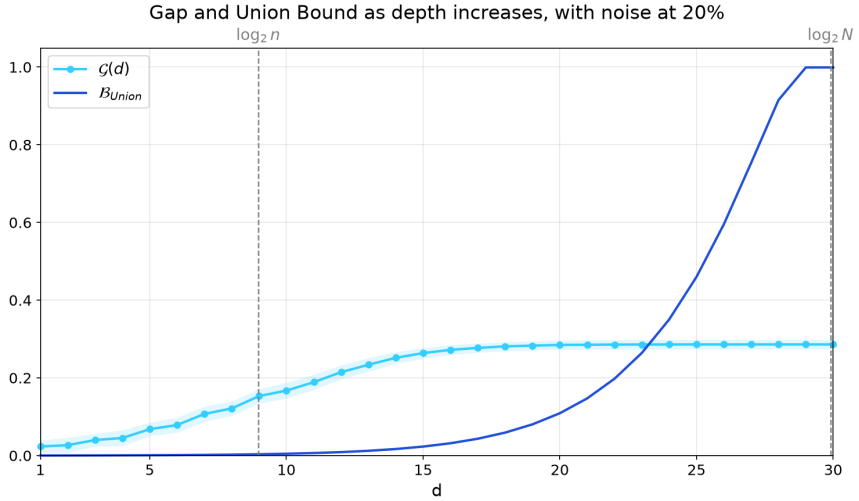


FIGURE 2

We plot the Union bound (11) against empirical curves in Figures 2 and 3 together with the empirical generalization gap $\mathcal{G}(d)$. The first thing catching the eye is that while the generalization gap saturates from depth 20, the union bound increases exponentially in d until depth 29. The saturating depth is only determined by N and is given by roughly $\log_2 N$. This is shown by equation (10) or Fact 3.2, combined with the fact that the Union bound depends on d only through $|\mathscr{H}_d|$.

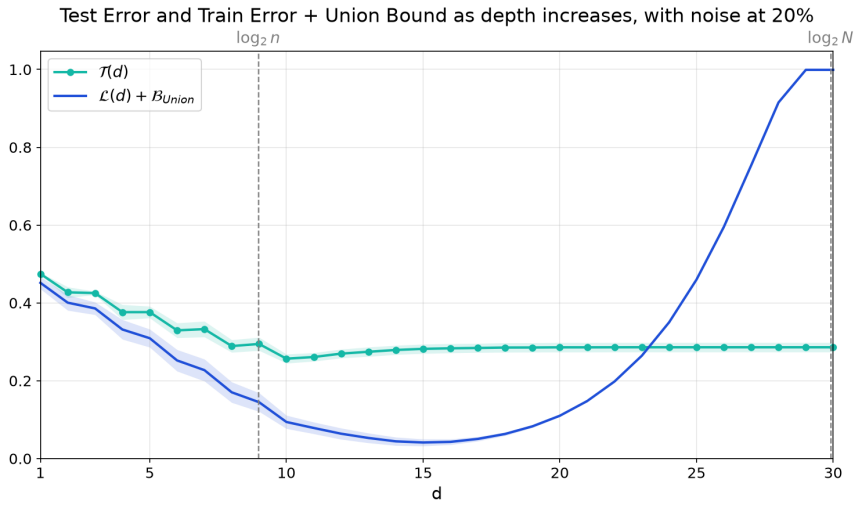


FIGURE 3

Note, however, that we can choose N arbitrarily large, and we use $N < \infty$ only for practical reasons. Therefore, the bound keeps increasing up to an arbitrarily large depth.

The train and test error of binary decision trees saturate because of a specific characteristic of the algorithm. As soon as the learned tree has only "pure" leaves, i.e. leaves containing only inputs that have the same label in the training set, the algorithm ceases to split them and hence any additional allowed depth does not change the learned model. Clearly, the cardinality of the hypothesis classes $|\mathscr{H}_d|$ does not capture this property.

Instead, a way of counting that accounts for the finiteness of our algorithm might reproduce the saturation of the test error curve. Specifically, the configuration with only pure leaves can be achieved for two reasons: either all the possible values that the inputs can take are classified, or all the inputs in the sample are classified. Since in our case $n \ll N$, the second event happens much earlier than the first one. As a consequence, a complexity measure that counts the functions in $\mathscr{H}$ *after* the algorithm has seen the data, i.e. that counts $h(x_1), \ldots, h(x_n) : h \in \mathscr{H}$, could more faithfully represent the true complexity of the space of models that the algorithm can practically learn and yield a bound that better captures the empirical test error curve.

4. Counting after Observing the Data

In this section, we introduce new tools that allow us to leverage the viewpoint exposed at the end of Section 3. This allows us to obtain a saturating bound on the generalization gap.

4.1. **Using the Rademacher Complexity.** First, we introduce the Rademacher complexity, a measure of the expressiveness of a class of functions. Then, we develop some results that build on it and yield the generalization bound.

Definition 4.1 (Empirical Rademacher complexity)**.** *Let $\xi = (\xi_1, \dots, \xi_n)$ be independent random variables taking value 1 or -1 with equal probability. Let $\mathcal{Z} \subset \mathbb{R}$ be a set and $\{z_1, \dots, z_n\} \subset \mathcal{Z}$ a collection of n points. Given $\mathscr{G} \coloneqq \{g : \mathcal{Z} \to \mathbb{R}\}$ a class of functions over $\mathcal{Z}$, the Rademacher complexity of $\mathscr{G}$ is*

$$\mathcal{R}_n(\mathscr{G}) \coloneqq \mathbb{E}\left[\sup_{g \in \mathscr{G}} \frac{1}{n} \sum_{i=1}^{n} \xi_i g(z_i)\right]$$

where the expectation is taken over the disorder of ξ.

Source. [17], Definition 3.1 (empirical Rademacher complexity).

The Rademacher complexity measures the richness of a class of functions. The argument of the supremum $\frac{1}{n}\sum_i \xi_i g(z_i)$ is the (scaled) inner product between $(\xi_1, \dots, \xi_n)$ and $(g(z_1), \dots, g(z_n))$, measuring how well functions in $\mathscr{G}$ can align to the Rademacher variables.

In the context of our learning problem, we look at the Rademacher complexity of the class $\mathscr{L} = \{\ell_h : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^+ : \ell_h(x, y) = \ell(h(x), y), h \in \mathscr{H}\}$.

Definition 4.2 (Rademacher complexity of $\mathscr{L}$)**.** *Let $\xi = (\xi_1, \dots, \xi_n)$ be independent random variables taking value 1 or -1 with equal probability. The Rademacher complexity of the class $\mathscr{L}$ is*

$$\mathcal{R}_n(\mathscr{L}) \coloneqq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i)\right]$$

where the expectation is taken over the disorder of ξ and D.

Source. [17], Definition 3.2 (Rademacher complexity), applied to the loss class $\mathscr{L}$.

Note that in this second definition we also take the expectation over the disorder of the data D, because our sample $(X_1, Y_1), \ldots, (X_n, Y_n)$ is random and our quantity of interest (3) is independent of the data realization. In this case, the argument of the supremum is the scaled inner product between the Rademacher vectors $(\xi_1, \ldots, \xi_n)$ and the loss vectors $(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n))$. Since the loss function is fixed, it measures to what extent classifiers $h \in \mathscr{H}$ can align to random noise. Note that the supremum is taken before the expectation over the disorder of ξ and D, and that it is a maximum, since $\mathscr{L}$ is finite. Hence, for each dataset and for each realization of the Rademacher vectors, we can choose a different function $\ell_h \in \mathscr{L}$ that attains the maximum. A high complexity means that for most values of ξ we can find a model h that is similar to that pattern. It is highest when we can find a function that correlates with any possible pattern.

The following results hold.

Theorem 4.3 (Rademacher complexity bound)**.** *[*[2]*, Proposition 4.2]*

$$\mathbb{E}\left[\sup_{h \in \mathscr{H}} (R - \hat{R}_D)(h)\right] \leq 2\mathcal{R}_n(\mathscr{L})$$

Source. The statement and the symmetrization argument are those of [2], Proposition 4.2. See also [17], Theorem 3.3, for the high-probability version. Here the Rademacher complexity is also averaged over the data, as in Definition 4.2.

Proof of Theorem 4.3. Let $Z \coloneqq (X, Y)$ and $Z_i \coloneqq (X_i, Y_i)$, so that

$$R(h) = \mathbb{E}[\ell_h(Z)] \qquad \text{and} \qquad \hat{R}_D(h) = \frac{1}{n}\sum_{i=1}^{n} \ell_h(Z_i).$$

Introduce an independent copy $D' = (Z'_1, \ldots, Z'_n)$ of the dataset D. Since the Z'_i are distributed as Z and are independent of D, we rewrite

$$R(h) = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i) \,\middle|\, D\right].$$

Then,

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] = \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} (\ell_h(Z'_i) - \ell_h(Z_i)) \,\middle|\, D\right]\right].$$

since $\ell_h(Z_i)$ is a function of D, hence $\mathbb{E}\left[\ell_h(Z_i) \mid D\right] = \ell_h(Z_i)$. Since the supremum of expectations is at most the expectation of the supremum, and by the law of total expectation, we obtain

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] \leq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right]. \tag{12}$$

Now we apply the symmetrization trick. Let $\mathscr{H} = \{h_1, \ldots, h_m\}$, with $m = |\mathscr{H}|$. Since Z_i and Z_i' are i.i.d. for each $i = 1, \ldots, n$, swapping them does not change the joint distribution, i.e.

$$\left(\ell_{h_j}(Z_i') - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \stackrel{d}{=} \left(\ell_{h_j}(Z_i) - \ell_{h_j}(Z_i')\right)_{j=1}^{m}.$$

This means that, for any fixed $(e_1, \ldots, e_n) \in \{-1, 1\}^n$,

$$\left(\ell_{h_j}(Z_i') - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \stackrel{d}{=} \left(e_i \left(\ell_{h_j}(Z_i') - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

The pairs $(Z_1, Z_1'), \ldots, (Z_n, Z_n')$ are also independent, therefore equality in distribution holds also when taking the sum, i.e.

$$\left(\frac{1}{n} \sum_{i=1}^{n} \left(\ell_{h_j}(Z_i') - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m} \stackrel{d}{=} \left(\frac{1}{n} \sum_{i=1}^{n} e_i \left(\ell_{h_j}(Z_i') - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

Finally, because the m random variables are jointly equally distributed, we have equality in distribution of the suprema

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right) \stackrel{d}{=} \sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} e_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right).$$

Because the two sides in the previous equation have the same distribution, their expected value is the same

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} e_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right].$$

Note that until now we have considered only the disorder of the data D, D'. Because this holds for any value of $(e_1, \ldots, e_n) \in \{-1, 1\}^n$, it holds surely for the random variable $\xi = (\xi_1, \ldots, \xi_n)$, i.e.

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right) \,\middle|\, \xi\right]$$

surely. Taking the expected value over the disorder of ξ, we obtain

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right] = \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right]$$

where the left hand side is independent of ξ and in the right hand side we use the law of total expectation. Plugging the result into (12), we have

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] \leq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right]$$

Now we split the sum over $i = 1, \ldots, n$ into

$$\frac{1}{n} \sum_{i=1}^{n} \xi_i \left(\ell_h(Z_i') - \ell_h(Z_i)\right) = \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i') + \frac{1}{n} \sum_{i=1}^{n} \left(-\xi_i \ell_h(Z_i)\right)$$

Then, we use that the supremum of the sum is bounded by the sum of the suprema,

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i') + \frac{1}{n} \sum_{i=1}^{n} \left(-\xi_i \ell_h(Z_i)\right) \leq \sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i') + \sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \left(-\xi_i \ell_h(Z_i)\right)$$

and that the expectation of the sum is the sum of the expectations, obtaining

$$\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] \leq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i')\right] + \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} (-\xi_i) \ell_h(Z_i)\right]$$

We also observe that we can replace $-\xi_i$ with ξ_i, because they have the same distributions, hence the same expected value, and are independent of the datasets. Therefore,

$$\begin{aligned}\mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] &\leq \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i')\right] + \mathbb{E}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(Z_i)\right] \\ &= 2\mathcal{R}_n(\mathscr{L}).\end{aligned}$$

□

Lemma 4.4 (Massart)**.** *[[17], Theorem 3.7] Let $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$ be a loss function bounded by $M \in \mathbb{R}_+$. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq M \sqrt{\frac{2 \log |\mathscr{L}|}{n}}.$$

Source. Adapted from [17], Theorem 3.7 (Massart's lemma), which is stated for a finite set of vectors $A \subset \mathbb{R}^n$ with $\max_{a \in A} \|a\|_2 \leq r$. We apply it to the loss class, where $r = M\sqrt{n}$, and we rewrite the proof with Hoeffding's lemma.

Proof of Lemma 4.4. For any $t > 0$,

$$\begin{aligned}
\exp\left(t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) &\le \mathbb{E}\left[\exp\left(t\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&\le \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}\mathbb{E}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\,\middle|\,D\right]\right] \quad (13)
\end{aligned}$$

where we use Jensen's inequality for the convex function $\exp(\cdot)$ in the first inequality, that exp is increasing in the first equality, we bound the supremum with the sum (all terms are positive) in the second inequality, and we use the independence of the ξ_i's, and their independence of D, in the last equality.

Now we apply Hoeffding's lemma (Theorem 2.1) to each factor for fixed $(X_i, Y_i) = (x_i, y_i)$ taking any value in $\mathcal{X}\times\mathcal{Y}$. The random variable $\xi_i\ell_h(x_i,y_i)$ is centered, since $\mathbb{E}[\xi_i] = 0$, and it satisfies $|\xi_i\ell_h(x_i,y_i)| = \ell_h(x_i,y_i) \le M$ by non-negativity of the loss and Proposition 2.12, hence it satisfies the assumptions of Hoeffding's lemma. We obtain

$$\mathbb{E}\left[\exp\left(t\xi_i\ell_h(x_i,y_i)\right)\right] \le e^{t^2M^2/2}.$$

Because this holds for any $(x_i,y_i)\in\mathcal{X}\times\mathcal{Y}$, we have

$$\mathbb{E}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\,\middle|\,D\right] \le e^{t^2M^2/2}$$

surely. Plugging this into (13),

$$\exp\left(t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) \le \mathbb{E}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}e^{t^2M^2/2}\right] = |\mathscr{L}|\,e^{nt^2M^2/2}.$$

Taking the logarithm of both sides,

$$t\,\mathbb{E}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right] \le \log|\mathscr{L}| + \frac{nt^2M^2}{2},$$

and dividing by $tn > 0$,

$$\mathcal{R}_n(\mathscr{L}) \le \frac{\log|\mathscr{L}|}{tn} + \frac{tM^2}{2} =: b(t) \quad (14)$$

Since (14) holds for every $t > 0$, we look for the tightest bound. We set the derivative in t to zero

$$\frac{\mathrm{d}b(t)}{\mathrm{d}t} = -\frac{\log|\mathscr{L}|}{t^2 n} + \frac{M^2}{2} = 0 \quad \Longrightarrow \quad t^\star = \sqrt{\frac{2\log|\mathscr{L}|}{nM^2}}.$$

Observe that since it is fair to assume $|\mathscr{L}| > 1$, the second derivative is nonnegative for any t

$$\frac{2\log|\mathscr{L}|}{t^3 n} \geq 0 \quad \forall t$$

hence $t^\star$ is a minimizer (if $|\mathscr{L}| = 1$, then $\log|\mathscr{L}| = 0$ and $b(t) = tM^2/2$ is increasing, so the bound follows by letting $t \to 0^+$). Plugging $t^\star$ into (14) yields

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log|\mathscr{L}|}{n}}$$

□

Note that Theorem 4.3 can be easily extended to a bound on the gap at $\hat{h}_D$ because

$$\mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq \mathbb{E}\left[\sup_{h\in\mathscr{H}}(R - \hat{R}_D)(h)\right]$$

since $\hat{h}_D \in \mathscr{H}$.

Note that Lemma 4.4 and Theorem 3.1 provide bounds with the same scaling law in $|\mathscr{L}|$ and $|\mathscr{H}|$ respectively.

Fact 4.5. *We have $|\mathscr{L}| \leq |\mathscr{H}|$ by definition of $\mathscr{L}$. Moreover, if ℓ is the zero-one loss, the inequality is an equality, as ℓ_h determines h.*

Thus, for the zero-one loss, Theorem 4.3 and Lemma 4.4 yield the same generalization bound as Theorem 3.1, up to constants. Now, we introduce a tool that allows us to take a step forward and improve this result.

Let

$$\mathscr{L}(D) \coloneqq \{(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) : h \in \mathscr{H}, D = ((X_i, Y_i))_{i=1}^n\}$$

be the set of loss vectors for a random dataset D, i.e. the set containing the functions in $\mathscr{L}$ applied to D. Let $v_i = \ell_h(X_i, Y_i)$ to ease notation.

The supremum in Definition 4.2 can be equivalently computed over the set of loss vectors,

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_i \xi_i \ell_h(X_i, Y_i) = \sup_{(v_1,\dots,v_n)\in \mathscr{L}(D)} \frac{1}{n} \sum_i \xi_i v_i$$

This yields the following version of the lemma.

Lemma 4.6 (Massart, loss vectors)**.** *Let $\ell : \mathcal{Y}\times\mathcal{Y} \to \mathbb{R}_+$ be a loss function bounded by $M \in \mathbb{R}_+$. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log \mathbb{E}[N_D(\mathscr{H})]}{n}}$$

where

$$N_D(\mathscr{H}) \coloneqq |\{(\ell_h(X_1, Y_1), \dots, \ell_h(X_n, Y_n)) : h \in \mathscr{H}, D = ((X_i, Y_i))_{i=1}^n\}| = |\mathscr{L}(D)|$$

is the cardinality of the set of loss vectors for the hypotheses in $\mathscr{H}$, which is a random variable as it depends on the dataset D.

Source. Same argument as Lemma 4.4, with the set of loss vectors $\mathscr{L}(D)$ in place of $\mathscr{L}$, as in [17], Section 3.2 (Corollaries 3.8–3.9). The number of loss vectors is random, hence we average it over D.

Proof of Lemma 4.6. Most of the proof is similar to Proof of Lemma 4.4.

Let $(\ell_h(X_1, Y_1), \dots, \ell_h(X_n, Y_n)) = \ell(D)$. For any $t > 0$,

$$\exp\left(t\,\mathbb{E}\left[\sup_{\ell(D)\in\mathscr{L}(D)} \sum_{i=1}^n \xi_i \ell_h(X_i, Y_i)\right]\right) \leq \mathbb{E}\left[\sum_{\ell(D)\in\mathscr{L}(D)} \prod_{i=1}^n \mathbb{E}\left[\exp\left(t\xi_i \ell_h(X_i, Y_i)\right) \,\middle|\, D\right]\right]$$

following the same steps as in the proof of Lemma 4.4, all of which are independent of the data disorder, as they can be made fixing an observation S.

Now we apply Hoeffding's lemma (Theorem 2.1). These steps also regard the argument of the expectation over the Rademacher variables disorder. We obtain

$$\mathbb{E}\left[\exp\left(t\xi_i \ell_h(X_i, Y_i)\right) \,\middle|\, D\right] \leq e^{t^2 M^2/2}.$$

surely as in Proof of Lemma 4.4. Plugging this into (13),

$$\begin{aligned}\exp\left(t\,\mathbb{E}\left[\sup_{\ell(D)\in\mathscr{L}(D)}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) &\leq \mathbb{E}\left[\sum_{\ell(D)\in\mathscr{L}(D)}\prod_{i=1}^{n}e^{t^2M^2/2}\right]\\ &= \mathbb{E}\left[N_D(\mathscr{H})e^{nt^2M^2/2}\right]\\ &= \mathbb{E}\left[N_D(\mathscr{H})\right]e^{nt^2M^2/2}.\end{aligned}$$

The rest of the proof follows exactly as Proof of Lemma 4.4, replacing $|\mathscr{L}|$ with $\mathbb{E}\left[N_D(\mathscr{H})\right]$. □

So, using the Rademacher complexity as an intermediate step allows us to replace the number of predictors $|\mathscr{H}|$ from Theorem 3.1 with the expected number of loss vectors $N_D(\mathscr{H})$ (up to constants). In fact, Theorem 4.3 bounds a quantity that depends on the true expected value of $\ell_h(X,Y)$ with a quantity that can be computed on loss vectors. The key ingredient that allows this is a symmetrization technique, which consists in replacing the deviation of the random variable $\ell_h(X,Y)$ from its expected value with the difference between two random variables $\ell_h(X,Y)$ and $\ell_h(X',Y')$ taking two i.i.d. copies D and D' as inputs. Note that since the loss function is fixed, counting loss vectors is equivalent to counting classification vectors. We work with loss vectors here because they arise more naturally from the Rademacher complexity, and turn to classification vectors in Section 4.2.

The following result contains the key point of the saturation mechanism discussed at the end of Section 3.

Lemma 4.7 (Counting)**.** *Let ℓ be the zero-one loss. Then*

$$N_D(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely. Source. Elementary counting step, own formulation; it is the idea behind the growth-function bounds in [17], Section 3.2.

Proof of Lemma 4.7. We prove that $N_S(\mathscr{H}) \leq |\mathscr{H}|$ and $N_S(\mathscr{H}) \leq 2^n$ for every $S \in (\mathcal{X}\times\mathcal{Y})^n$. Fix $S = (z_1,\ldots,z_n)$ and consider the map

$$\Phi\colon \mathscr{H} \to \{0,1\}^n, \qquad h \mapsto \big(\ell_h(z_1),\ldots,\ell_h(z_n)\big),$$

so that $N_S(\mathscr{H}) = |\Phi(\mathscr{H})|$. Since the image of a set under any map has cardinality at most that of the domain, $|\Phi(\mathscr{H})| \leq |\mathscr{H}|$. Since $\Phi(\mathscr{H}) \subseteq \{0,1\}^n$, we also have $|\Phi(\mathscr{H})| \leq |\{0,1\}^n| = 2^n$. Because this holds for every realization S of D, the claim for $N_D(\mathscr{H})$ follows. □

All in all, we obtain the following bound on the generalization gap.

Theorem 4.8 (Rademacher and Massart bound)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}[(R - \hat{R}_D)(\hat{h}_D)] \leq \min\left\{2\sqrt{\frac{2\log|\mathscr{H}|}{n}}, 2\sqrt{2\log 2}\right\}$$

Source. Own combination of Theorem 4.3, Lemma 4.6 and Lemma 4.7. A high-probability analogue is [17], Corollaries 3.8–3.9.

Proof of Theorem 4.8. Since $\hat{h}_S \in \mathscr{H}$ for any realization S of D,

$$\mathbb{E}\left[(R - \hat{R}_D)(\hat{h}_D)\right] \leq \mathbb{E}\left[\sup_{h\in\mathscr{H}}(R - \hat{R}_D)(h)\right]$$

The proof is concluded applying Theorem 4.3, bounding the Rademacher complexity with Lemma 4.6, and bounding $N_D(\mathscr{H})$ with Lemma 4.7. □

We now plot the obtained bound in the context of decision trees to compare it with the empirical gap curve.

Experimental design

We use a heuristic normalization of the obtained bound and call this the *Rademacher-Massart bound*,

$$\mathcal{B}_{RadMass}(d) = C_{RadMass} \cdot \min\left\{2\sqrt{\frac{2\log|\mathscr{H}_d|}{n}}, 2\sqrt{2\log 2}\right\} \tag{15}$$

where $C_{RadMass} = 0.125$ so that the curve is of the same scale as the generalization gap. See Appendix B for details on how this constant is fitted. The rest of the experimental design is as above.

Empirical findings

We plot our newly found bound, equation (15), against the Union bound from Section 3 in Figure 4. We notice immediately that the Rademacher-Massart bound

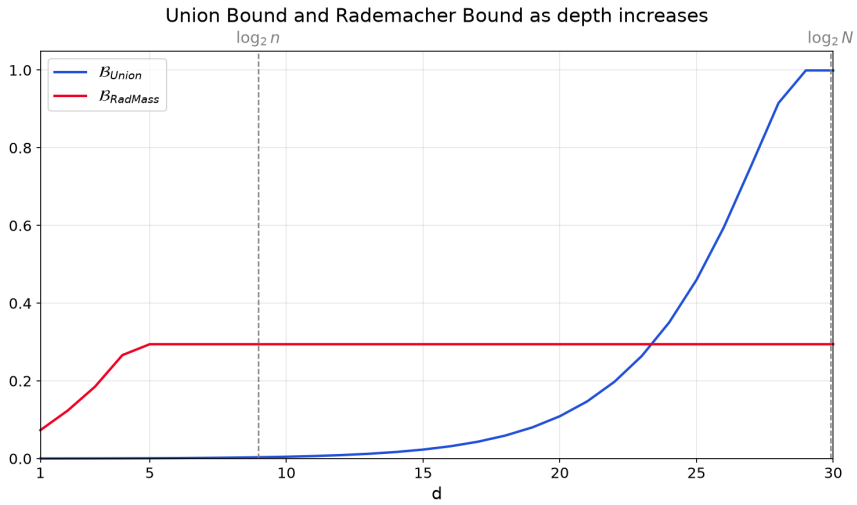


FIGURE 4

better captures the curve of the test error, because it saturates from a lower depth. Specifically, for low values of the depth d, the minimum in Lemma 4.7 is attained by $|\mathscr{H}_d|$, while for a large enough depth the minimum becomes 2^n and the curve saturates. The depth at which the bound saturates is given by

$$\log|\mathscr{H}_d| \approx (2^d - 1)\log\frac{eN}{2^d} \geq n\log 2 \quad \Rightarrow \quad d \geq 5$$

for a fixed $n = 500$.

Next, we compare the Rademacher-Massart bound with empirical error curves in Figures 5 and 6. As expected, we notice that the bound captures the saturation of the generalization gap (or test error). At the same time, it does not capture its shape in terms of the saturating depth and the initial steepness. We suppose that the different saturating depth between the two curves is due to a double effect:

- the empirical curve saturates well beyond the theoretical interpolation threshold $\log_2 n$, due to the greediness of the algorithm used in the experiments;
- the bound curve saturates at a depth that depends on N, which as discussed above could be arbitrarily big: the larger N, the earlier the saturation.

We leave this as a direction for future work.

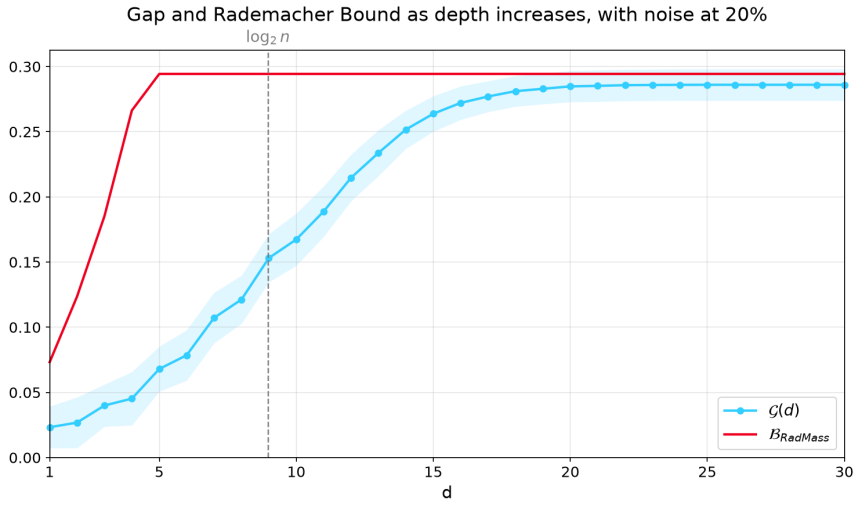


FIGURE 5

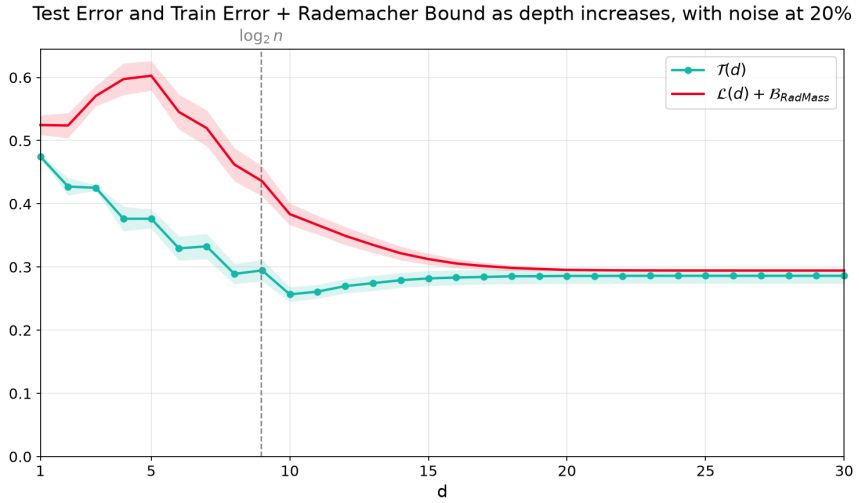


FIGURE 6

4.2. **Turning to the Growth Function.** Let us introduce the growth function, a new measure of complexity of the hypothesis class that considers classification vectors instead of loss vectors. This tool allows us to make the intuition on the saturation of the test error due to characteristics of the decision tree algorithm

that we discussed in Section 4.1 more explicit, as we count directly the distinct hypotheses in $\mathscr{H}$ after the algorithm has seen the data, instead of their loss.

Definition 4.9 (Growth function of $\mathscr{H}$)**.** *The growth function of $\mathscr{H}$ is*

$$\Pi_n(\mathscr{H}) := \sup_{x_1,\dots,x_n \in \mathcal{X}} \left|\{(h(x_1),\dots,h(x_n)) : h \in \mathscr{H}\}\right|.$$

Source. [17], Definition 3.3 (growth function).

The following extends Lemma 4.7.

Lemma 4.10 (Counting with growth function)**.** *Let ℓ be the zero-one loss. Then*

$$1 \le N_D(\mathscr{H}) \le \Pi_n(\mathscr{H}) \le \min\{|\mathscr{H}|, 2^n\}$$

surely. Moreover,

$$\Pi_n(\mathscr{H}) = N_n(\mathscr{H}) \coloneqq \sup_{S \in (\mathcal{X}\times\mathcal{Y})^n} N_S(\mathscr{H})$$

where

$$N_S(\mathscr{H}) := |\{(\ell_h(x_1,y_1),\dots,\ell_h(x_n,y_n)) : h \in \mathscr{H}, S = ((x_i,y_i))_{i=1}^n\}|$$

is a realization of $N_D(\mathscr{H})$ for a fixed dataset S.

Source. Own extension of Lemma 4.7, relating the number of loss vectors to the growth function of [17], Section 3.2.

Proof of Lemma 4.10. We prove the inequalities for $N_S(\mathscr{H})$, for every $S \in (\mathcal{X} \times \mathcal{Y})^n$. Since $N_D(\mathscr{H})$ takes the value $N_S(\mathscr{H})$ when D takes the value S, they hold for $N_D(\mathscr{H})$ for every realization of D.

Fix $S = ((x_1,y_1),\dots,(x_n,y_n)) \in (\mathcal{X}\times\mathcal{Y})^n$, and consider the maps

$$\Psi\colon \mathscr{H} \to \mathcal{Y}^n, \quad h \mapsto \big(h(x_1),\dots,h(x_n)\big),$$

$$\lambda\colon \mathcal{Y}^n \to \{0,1\}^n, \quad (\hat{y}_1,\dots,\hat{y}_n) \mapsto \big(\ell(\hat{y}_1,y_1),\dots,\ell(\hat{y}_n,y_n)\big).$$

The map $\Phi\colon h \mapsto (\ell_h(x_1,y_1),\dots,\ell_h(x_n,y_n))$ of the proof of Lemma 4.7 factorizes as $\Phi = \lambda \circ \Psi$, hence $N_S(\mathscr{H}) = |\Phi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))|$. Moreover, by definition of growth function, $|\Psi(\mathscr{H})| \le \Pi_n(\mathscr{H})$.

$1 \leq N_S(\mathscr{H})$. Since $\mathscr{H}$ is nonempty, $\Phi(\mathscr{H})$ is nonempty.

$N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. Since the image of a set under any map has cardinality at most that of the set,

$$N_S(\mathscr{H}) = |\lambda(\Psi(\mathscr{H}))| \leq |\Psi(\mathscr{H})| \leq \Pi_n(\mathscr{H}).$$

$\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$. Let $x_1, \ldots, x_n \in \mathcal{X}$ and let Ψ be the corresponding map. As above, $|\Psi(\mathscr{H})| \leq |\mathscr{H}|$, and $\Psi(\mathscr{H}) \subseteq \mathcal{Y}^n$ with $|\mathcal{Y}^n| = 2^n$, since $\mathcal{Y} = \{-1, 1\}$. Taking the supremum over $x_1, \ldots, x_n$ gives $\Pi_n(\mathscr{H}) \leq |\mathscr{H}|$ and $\Pi_n(\mathscr{H}) \leq 2^n$.

$\Pi_n(\mathscr{H}) = N_n(\mathscr{H})$. Taking the supremum over S in $N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$ gives $N_n(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. For the converse, let $x_1, \ldots, x_n \in \mathcal{X}$, take any $y_1, \ldots, y_n \in \mathcal{Y}$, and let $S = ((x_1, y_1), \ldots, (x_n, y_n))$. Since $\mathcal{Y} = \{-1, 1\}$ and ℓ is the zero-one loss, $\hat{y}_i = y_i$ if $\ell(\hat{y}_i, y_i) = 0$ and $\hat{y}_i = -y_i$ if $\ell(\hat{y}_i, y_i) = 1$. Hence λ is injective, and

$$|\Psi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))| = N_S(\mathscr{H}) \leq N_n(\mathscr{H}).$$

Taking the supremum over $x_1, \ldots, x_n$ gives $\Pi_n(\mathscr{H}) \leq N_n(\mathscr{H})$. □

The relation between N_D and Π_n makes it immediate to obtain a growth-function version of Lemma 4.6 (substitute $N_D(\mathscr{H}) \leq \Pi_n(\mathscr{H})$). Substituting also $\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$ yields exactly the bounds of Theorem 4.8 and Theorem 5.5. We continue using this measure of complexity in the next section.

5. The Effect of Noise

In this section, we turn to analyzing the effect of the noise in the data on the generalization ability of the algorithm and we look for a bound that captures this effect.

The plots in Figures 7 and 8 show the generalization gap and the test error, respectively, for different levels of noise in the data.

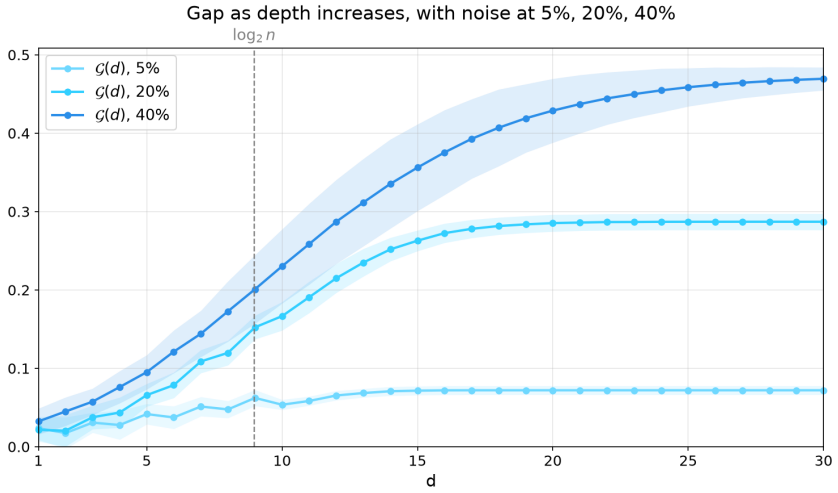


Figure 7

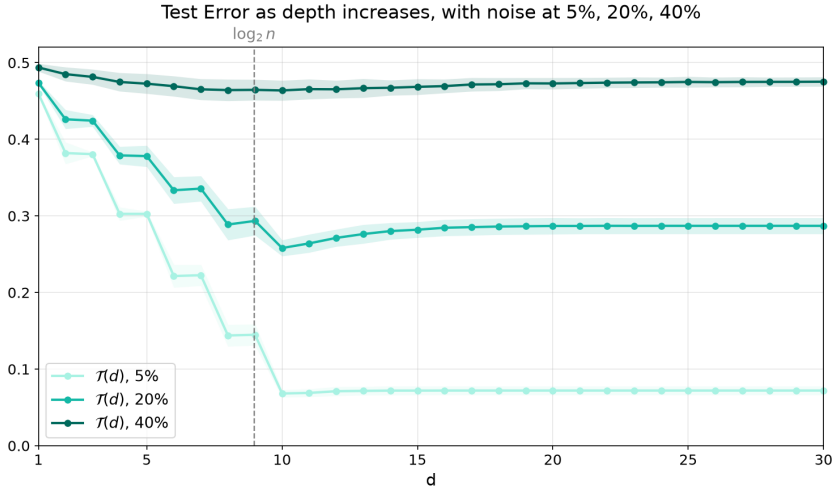


Figure 8

As noise increases, the test error curve tends to flatten at 0.5, the error corresponding to guessing perfectly random labels. As a consequence, also the generalization gap $\mathcal{G}(d) = \mathcal{T}(d) - \mathcal{L}(d)$ saturates at a higher level for higher levels of noise, as seen in Figure 7. Therefore, we would like our bound $\mathcal{G}(d) \lesssim \mathcal{B}(d)$ to similarly change for different levels of noise.

Figure 9 shows that the bound obtained in the previous section ($\mathcal{B}_{RadMass}$, equation (15)) fails to capture the sensitivity of the generalization gap to noise. Indeed, the orange, red, and bordeaux curves coincide and the plot shows only one of them, because the function does not change with different noise levels.

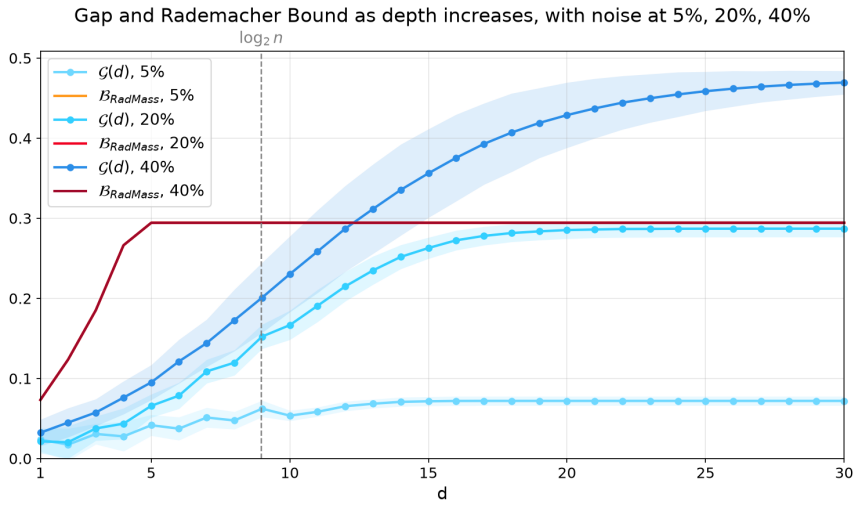


FIGURE 9

5.1. **Dependence on the Best Risk.** We implement a different approach and new theoretical tools to look for a noise-sensitive bound.

Let ℓ be the zero-one loss for the rest of this section.

Lemma 5.1 (Relative deviation)**.** *For $\varepsilon > 0$ with $n\varepsilon^2 \geq 8$,*

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \leq 2\Pi_{2n}(\mathscr{H})e^{-n\varepsilon^2/16}.$$

Source. Relative deviation inequality in the spirit of [7], Theorem 5.1, which relies on [4], Section 2, Theorem 1 (see also [1]). The statement is written for the zero-one loss with the growth function $\Pi_{2n}(\mathscr{H})$, and the constants are not optimized.

Proof of Lemma 5.1. Recall $\mathscr{L} = \{(x, y) \mapsto \ell(y, h(x)) : h \in \mathscr{H}\}$, where ℓ is the zero-one loss. Applying the first inequality of Theorem 5.1 of [7], with $\mathbb{S}_{\mathcal{F}}(X_1^{2n}) \leq N_{2n}(\mathscr{H}) = \Pi_{2n}(\mathscr{H})$ by Lemma 4.10, we obtain that with probability at least $1-\delta$, simultaneously for all $h \in \mathscr{H}$,

$$R(h) - \widehat{R}_D(h) \leq 2\sqrt{\frac{R(h)\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}}.$$

Set

$$\delta_\varepsilon = 4\Pi_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon < 1$, applying the preceding inequality with $\delta = \delta_\varepsilon$ gives

$$\Pr\left[\exists h \in \mathscr{H} : R(h) - \widehat{R}_D(h) > \varepsilon\sqrt{R(h)}\right] \leq 4\Pi_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon \geq 1$, this probability bound holds trivially. Finally, since $n\varepsilon^2 \geq 8$,

$$4e^{-n\varepsilon^2/4} \leq 2e^{-n\varepsilon^2/16},$$

which proves the claim. □

The result above is a concentration result for the random variable $\hat{R}_D(h)$ in terms of the true risk $R(h)$. The key ingredient in its proof is Chebyshev's inequality (Theorem 2.3) that bounds the deviation of a random variable in terms of its variance. This is a more refined result as compared to Hoeffding's inequality used in Theorem 3.1, which only uses boundedness of the random variable. Moreover, when a random variable is bounded, its variance is controlled by its expected risk because, intuitively, it cannot fluctuate much around its center. In particular, for $\ell_h(X, Y)$ that is bounded and Bernoulli, we have

$$\mathrm{Var}(\ell_h) = \mathbb{E}[\ell_h](1 - \mathbb{E}[\ell_h]) \leq \mathbb{E}[\ell_h] = R(h).$$

This explains how we get a bound on the deviation relative to the true risk. The bound we obtain, moreover, depends on the growth function $\Pi_{2n}(\mathscr{H})$, i.e. the maximum number of classification vectors for datasets of size $2n$. This quantity arises naturally because the result uses a symmetrization technique. Also, as presented in Section 4.1, this technique considers two independent datasets D and D' and we count loss vectors over their union, hence the size $2n$ instead of n.

In high probability, and isolating the empirical risk $\hat{R}_D$ on the right hand side, we obtain the following result.

Theorem 5.2 (Bound with empirical risk)**.** *With probability at least* $1-\delta$*,*

$$(R-\hat{R}_D)(\hat{h}_D) \le 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2\Pi_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2\Pi_{2n}(\mathscr{H})/\delta)}{n}$$

Source. Analogous to [7], Theorem 5.1, rewritten to isolate $\hat{R}_D(\hat{h}_D)$ by solving a quadratic inequality, starting from Lemma 5.1.

Proof of Theorem 5.2. Let $\delta \in (0,1)$ and set

$$u \coloneqq \frac{\log(2\Pi_{2n}(\mathscr{H})/\delta)}{n} \qquad \text{and} \qquad \varepsilon \coloneqq 4\sqrt{u},$$

so that $2\Pi_{2n}(\mathscr{H})e^{-n\varepsilon^2/16} = \delta$. Since $\Pi_{2n}(\mathscr{H}) \ge 1$ and $\delta < 1$, we have $n\varepsilon^2 = 16\log(2\Pi_{2n}(\mathscr{H})/\delta) \ge 16\log 2 \ge 8$, hence Lemma 5.1 applies and gives

$$\Pr\left[\exists h \in \mathscr{H} : (R-\hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \le \delta.$$

Therefore, with probability at least $1-\delta$, we have $(R-\hat{R}_D)(h) \le \varepsilon\sqrt{R(h)}$ for all $h \in \mathscr{H}$, and in particular for $h = \hat{h}_D \in \mathscr{H}$.

On this event, we isolate $\hat{R}_D(\hat{h}_D)$. Let $r \coloneqq R(\hat{h}_D)$ and $\hat{r} \coloneqq \hat{R}_D(\hat{h}_D)$, so that $r - \varepsilon\sqrt{r} - \hat{r} \le 0$. This is a quadratic inequality in $x \coloneqq \sqrt{r} \ge 0$, namely $x^2 - \varepsilon x - \hat{r} \le 0$, which implies that x is at most the largest root,

$$\sqrt{r} \le \frac{\varepsilon + \sqrt{\varepsilon^2 + 4\hat{r}}}{2}.$$

Squaring both sides,

$$r \le \frac{2\varepsilon^2 + 4\hat{r} + 2\varepsilon\sqrt{\varepsilon^2+4\hat{r}}}{4} = \hat{r} + \frac{\varepsilon^2}{2} + \frac{\varepsilon}{2}\sqrt{\varepsilon^2+4\hat{r}} \le \hat{r} + \varepsilon^2 + \varepsilon\sqrt{\hat{r}},$$

where in the last inequality we used $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. Hence, substituting $\varepsilon = 4\sqrt{u}$, with probability at least $1-\delta$,

$$\begin{aligned}(R-\hat{R}_D)(\hat{h}_D) = r - \hat{r} &\le \varepsilon\sqrt{\hat{r}} + \varepsilon^2 \\ &= 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2\Pi_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2\Pi_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

We now want to bound the empirical risk of the ERM $\hat{R}_D(\hat{h}_D)$ in the right-hand side of the result above.

Let

$$R^*_{\mathscr{H}} = \min_{h \in \mathscr{H}} R(h)$$

be the minimum risk attainable in the hypothesis class and let

$$h_b \in \arg\min_{h \in \mathscr{H}} R(h)$$

be the best hypothesis, i.e. the one attaining $R^*_{\mathscr{H}}$. Again, at least one such model exists since $\mathscr{H}$ is finite and nonempty.

By Bernstein's inequality (Theorem 2.4) on $\hat{R}_D(h_b)$ and the definition of ERM (equation (2)), we obtain the following result.

Corollary 5.3 (Bound with best risk)**.** *With probability at least* $1 - \delta$*,*

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{R^*_{\mathscr{H}} \log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}$$

Source. Own result; for a related ERM excess-risk bound depending on the best risk in the class see [4], Section 3, Theorem 2. Here we bound instead the generalization gap, using Bernstein's inequality on $\hat{R}_D(h_b)$ and Theorem 5.2.

Proof of Corollary 5.3. Recall $h_b = \arg\min_{h \in \mathscr{H}} R(h)$, $R(h_b) = R^*_{\mathscr{H}}$ by definition of $R^*_{\mathscr{H}}$ and $\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_b)$ by definition of the ERM $\hat{h}_D$. Since h_b is independent of the data, the variables $\ell(h_b(X_i), Y_i)$ are i.i.d. Bernoulli with mean $R^*_{\mathscr{H}}$ and variance $R^*_{\mathscr{H}}(1 - R^*_{\mathscr{H}}) \leq R^*_{\mathscr{H}}$. Bernstein's inequality (Theorem 2.4) with $M = 1$ gives, with probability at least $1 - \delta/2$,

$$\hat{R}_D(h_b) \leq R^*_{\mathscr{H}} + \sqrt{\frac{2R^*_{\mathscr{H}} \log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

Let $u \coloneqq \log(4\Pi_{2n}(\mathscr{H})/\delta)/n$. Since $\Pi_{2n}(\mathscr{H}) \geq 1$, $\log(2/\delta)/n \leq u$. Thus

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_b) \leq R^*_{\mathscr{H}} + 2\sqrt{R^*_{\mathscr{H}} u} + u = \left(\sqrt{R^*_{\mathscr{H}}} + \sqrt{u}\right)^2. \tag{16}$$

Theorem 5.2 at confidence $1 - \delta/2$ gives, with probability at least $1 - \delta/2$,

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\hat{R}_D(\hat{h}_D)\, u} + 16u. \tag{17}$$

By a union bound, both events hold simultaneously with probability at least $1-\delta$. Hence, plugging (16) in the RHS of (17), we obtain that with probability at least $1-\delta$,

$$\begin{aligned}(R-\hat{R}_D)(\hat{h}_D) &\leq 4\sqrt{u}\left(\sqrt{R^*_{\mathscr{H}}}+\sqrt{u}\right)+16u \\ &= 4\sqrt{\frac{R^*_{\mathscr{H}}\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}}+20\frac{\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

The result above yields a bound on the generalization gap that depends on the best risk. The key to obtaining this dependence is that Bernstein's is a concentration result in terms of the variance, like Chebyshev's, and $\mathrm{Var}(\ell_{h_b}) \leq \mathbb{E}[\ell_{h_b}] = R^*_{\mathscr{H}}$.

We now make the dependence on the noise explicit.

Let

$$R^* := \inf_{h:\mathcal{X}\to\mathcal{Y}} R(h)$$

be the Bayes risk. Note that the infimum in the definition runs over a superset of $\mathscr{H}$, hence $R^* \leq R^*_{\mathscr{H}}$. The best risk can be decomposed as

$$R^*_{\mathscr{H}} = \underbrace{R^*}_{\text{Bayes risk}} + \underbrace{R^*_{\mathscr{H}} - R^*}_{\text{Approximation error}} =: R^* + A_{\mathscr{H}}$$

where

- the Bayes risk R^* measures the classification error due to the irreducible noise ε, as it is the minimal error that any function $h : \mathcal{X} \to \mathcal{Y}$ necessarily makes;
- the approximation error $A_{\mathscr{H}}$ is the excess error we make given that we consider a restricted set of predictors, $\mathscr{H}$.

Note that if there is no noise, then $R^* = 0$ and the Bayes risk is attained by the sign of the data-generating function $\mathrm{sign} \circ f^*$. Moreover, if the model attaining R^* lives in the hypothesis class, then $A_{\mathscr{H}} = 0$.

We obtain the following result.

Corollary 5.4. *With probability* $1-\delta$,

$$(R-\hat{R}_D)(\hat{h}_D) \le 4\sqrt{\frac{(R^*+A_{\mathscr{H}})\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}$$

Source. Own result: it follows from Corollary 5.3 and the decomposition of the best risk into Bayes risk (see [7], Section 2) and approximation error.

Proof of Corollary 5.4. Substitute $R^*_{\mathscr{H}} = R^* + A_{\mathscr{H}}$ into Corollary 5.3. □

Tail integrating and using the counting lemma (Lemma 4.10) yield the following theorem.

Theorem 5.5 (Noise and approximation error bound)**.** *Let* ℓ *be the zero-one loss. Then*

$$\mathbb{E}\left[(R-\hat{R}_D)(\hat{h}_D)\right] \le \min\left\{4\sqrt{\frac{(R^*+A_{\mathscr{H}})\,(\log(4|\mathscr{H}|)+1)}{n}} + 20\,\frac{\log(4|\mathscr{H}|)+1}{n},\right.$$
$$\left. 4\sqrt{\frac{(R^*+A_{\mathscr{H}})\,((2n+2)\log 2+1)}{n}} + 20\,\frac{(2n+2)\log 2+1}{n}\right\}$$

Source. Own result: it follows from Corollary 5.4 by tail integration (Proposition 2.6) and Lemma 4.10.

Proof of Theorem 5.5. Let $\text{gap} = (R-\hat{R}_D)(\hat{h}_D)$ and $C = \sqrt{\frac{\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n}}$. Corollary 5.3 gives

$$\Pr\left[\text{gap} \le 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2\right] \ge 1-\delta$$
$$\implies \Pr\left[\text{gap} > 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2\right] \le \delta$$

Set $\varepsilon = 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2+4\sqrt{R^*_{\mathscr{H}}}\,C-\varepsilon = 0 \implies C = \frac{-4\sqrt{R^*_{\mathscr{H}}}+\sqrt{16R^*_{\mathscr{H}}+80\varepsilon}}{40} = \frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}$$

So we have

$$\frac{\log(4\Pi_{2n}(\mathscr{H})/\delta)}{n} = \left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2$$
$$\frac{4\Pi_{2n}(\mathscr{H})}{\delta} = \exp\left(n\left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2\right)$$

and hence

$$\delta(\varepsilon) = 4\Pi_{2n}(\mathscr{H}) \exp\left(-\frac{n}{100}\left(\sqrt{R^*_{\mathscr{H}} + 5\varepsilon} - \sqrt{R^*_{\mathscr{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X] = \int_0^\infty \Pr[X \geq \varepsilon]\, d\varepsilon$ for $X \geq 0$, applied to $\mathrm{gap}^+ \coloneqq \max\{\mathrm{gap}, 0\}$,

$$\mathbb{E}[\mathrm{gap}] \leq \mathbb{E}[\mathrm{gap}^+] = \int_0^\infty \Pr[\mathrm{gap} \geq \varepsilon]\, d\varepsilon \leq \int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon.$$

We now compute the integral by exchanging the roles of ε and δ. The map $\varepsilon \mapsto \delta(\varepsilon)$ is continuous and strictly decreasing on $[0, \infty)$, with $\delta(0) = 4\Pi_{2n}(\mathscr{H}) > 1$ and $\delta(\varepsilon) \to 0$ as $\varepsilon \to \infty$. Hence, for every $t \in (0, 1)$ there is a unique $\varepsilon(t) > 0$ such that $\delta(\varepsilon(t)) = t$, and it is given by $\varepsilon(t) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ with $C = \sqrt{\log(4\Pi_{2n}(\mathscr{H})/t)/n}$, by construction of $\delta(\varepsilon)$. Moreover, for $t \in (0, 1)$,

$$\min\{1, \delta(\varepsilon)\} > t \iff \delta(\varepsilon) > t \iff \varepsilon < \varepsilon(t).$$

Writing $\min\{1, \delta(\varepsilon)\} = \int_0^1 \mathbf{1}\{t < \min\{1, \delta(\varepsilon)\}\}\, dt$ and exchanging the order of integration by Tonelli's theorem (the integrand is non-negative), we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \int_0^\infty \mathbf{1}\{\varepsilon < \varepsilon(t)\}\, d\varepsilon\, dt = \int_0^1 \varepsilon(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta.$$

Hence, writing $\varepsilon(\delta) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}[\mathrm{gap}] \leq \int_0^1 \varepsilon(\delta)\, d\delta = 4\sqrt{\frac{R^*_{\mathscr{H}}}{n}} \int_0^1 \sqrt{\log \frac{4\Pi_{2n}(\mathscr{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4\Pi_{2n}(\mathscr{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4\Pi_{2n}(\mathscr{H})}{\delta}\, d\delta = \log(4\Pi_{2n}(\mathscr{H})) - \int_0^1 \log \delta\, d\delta = \log(4\Pi_{2n}(\mathscr{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4\Pi_{2n}(\mathscr{H})}{\delta}}\, d\delta \leq \sqrt{\int_0^1 \log \frac{4\Pi_{2n}(\mathscr{H})}{\delta}\, d\delta} = \sqrt{\log(4\Pi_{2n}(\mathscr{H})) + 1},$$

we obtain

$$\mathbb{E}[\mathrm{gap}] \leq 4\sqrt{\frac{R^*_{\mathscr{H}}\left(\log(4\Pi_{2n}(\mathscr{H})) + 1\right)}{n}} + 20\, \frac{\log(4\Pi_{2n}(\mathscr{H})) + 1}{n}.$$

The right-hand side is increasing in $\Pi_{2n}(\mathscr{H})$. Hence, substituting $R^*_{\mathscr{H}} = R^* + A_{\mathscr{H}}$ and $\Pi_{2n}(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^{2n}\}$ (Lemma 4.10 with $2n$ in place of n), and noting that $\log(4 \cdot 2^{2n}) = (2n + 2)\log 2$, we obtain the claim.

Asymptotically,

$$\mathbb{E}\left[(R-\hat{R}_D)(\hat{h}_D)\right] \leq O\left(\sqrt{\frac{(R^*+A_{\mathscr{H}})\,\log \Pi_{2n}(\mathscr{H})}{n}}+\frac{\log \Pi_{2n}(\mathscr{H})}{n}\right)$$

because, at the numerator,

$$\log(4\Pi_{2n}(\mathscr{H}))+1=\log 4+\log \Pi_{2n}(\mathscr{H})+1=\log \Pi_{2n}(\mathscr{H})+O(1)$$

and for large x,

$$\log x+O(1)\leq C\log x$$

for some constant $C>0$. Applying $\Pi_{2n}(\mathscr{H})\leq \min\{|\mathscr{H}|, 2^{2n}\}$ (Lemma 4.10), we obtain

$$\mathbb{E}\left[(R-\hat{R}_D)(\hat{h}_D)\right] \leq \min\left\{O\left(\sqrt{\frac{(R^*+A_{\mathscr{H}})\log|\mathscr{H}|}{n}}+\frac{\log|\mathscr{H}|}{n}\right), O\left(\sqrt{R^*+A_{\mathscr{H}}}+1\right)\right\}$$

□

The result above is a bound on the expected gap at the ERM that explicitly depends on the noise R^* and that uses a measure of complexity that counts hypotheses after the algorithm has seen the data. We have obtained it by using more advanced concentration inequalities that take into account the variance of the random variable, while Hoeffding's inequality and Hoeffding's lemma used in Sections 3 and 4.1 are myopic to the distribution of the random variable and only use that it ranges over a bounded interval. We plot it and investigate how it relates to the empirical error curve of decision trees. Note that Corollary 5.3 uses the definition of ERM. Therefore, strictly speaking the bound obtained in Theorem 5.5 does not hold for the model selected by the DecisionTreeClassifier algorithm used in the experiments. Nonetheless, we present the comparison between the two curves for explorative purposes.

Experimental design

As compared to Sections 3 and 4.1 that used a Gaussian noise, we switch to a label-flip noise with flip probability $1-p$. This makes the computation of the Bayes risk and of the best risk easier. Therefore, we generate the labels as

$$y \leftarrow \text{sign}(f^*(x))\cdot b$$

where f^* is the sine function (9) and b is a random variable taking value 1 with probability p and -1 with probability $1-p$. We use $p \in \{0.6, 0.8, 0.95\}$.

Since $p \geq 0.5$, the Bayes risk is

$$R^* = 1 - p.$$

This is because the hypothesis attaining the Bayes risk selects the most probable label for each input (*Bayes classifier* in Section 2 of [7]). When $\Pr(\text{flip}) = 1 - p < 0.5$, the most probable label is the noiseless one determined by $\text{sign}(f^*(x))$, which gives error $1-p$. The best risk $R^*_{\mathscr{H}_d}$ is computed exactly for each d using dynamic programming (see Section B.3 in the Appendix for details on the computation of the best risk).

As always, we plot a normalized version of the bound obtained in Theorem 5.5, which we call *Noise bound*:

$$\begin{aligned}\mathcal{B}_{Noise}(d) = C_{Noise} \cdot \min \Bigg\{ & 4\sqrt{\frac{(R^*_{\mathscr{H}_d})\,(\log(4|\mathscr{H}|)+1)}{n}} + 20\,\frac{\log(4|\mathscr{H}|)+1}{n}, \\ & 4\sqrt{\frac{(R^*_{\mathscr{H}_d})\,((2n+2)\log 2+1)}{n}} + 20\,\frac{(2n+2)\log 2+1}{n} \Bigg\}\end{aligned} \tag{18}$$

with $C_{Noise} = 0.01$ so that the curve is of the same scale as the generalization gap. See Appendix B for details on how this constant is fitted.

Empirical findings

Figure 10 shows the Noise bound $\mathcal{B}_{Noise}$ from equation (18) against the empirical curve of the generalization gap. The curve is similar to that of the Rademacher-Massart bound (Figure 5), with the bound generally increasing at smaller depths before reaching a plateau. Hence, the Noise bound also captures the saturation of the gap, as expected thanks to the counting lemma. Moreover, Figure 11 shows that the bound moves for different levels of the noise, capturing the sensitivity of the empirical error that we have seen at the beginning of this section. The saturating value is higher for higher levels of noise, as happens to the generalization gap. At the same time, the change in the theoretical curve is much smaller than the empirical one, which constitutes a limitation to the descriptive power of this curve. Nonetheless, the Noise bound captures some aspects of the generalization ability

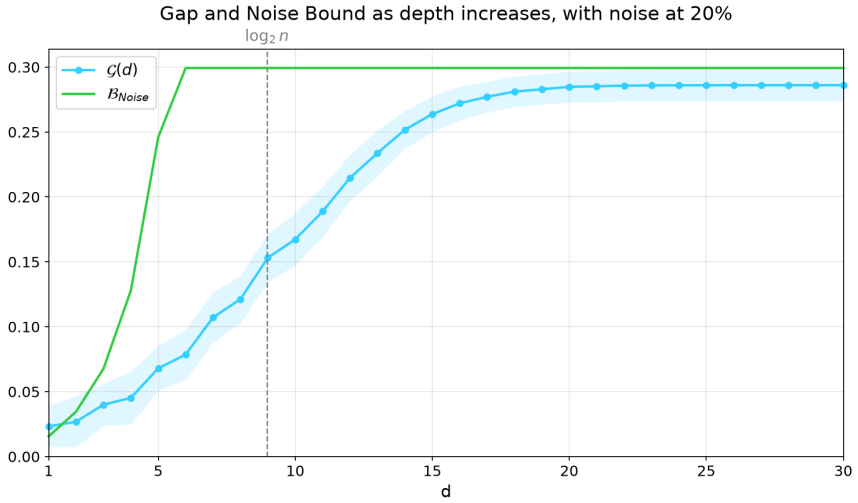


FIGURE 10

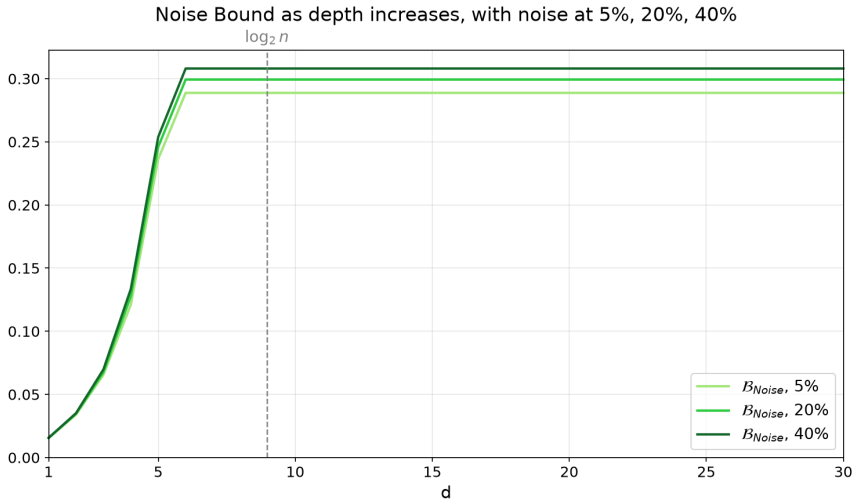


FIGURE 11

(its saturation and the effect of noise) through the dependence on the best risk and the counting lemma.

To understand better how the Noise bound depends on the noise, we look at its only noise-dependent ingredient, the best risk $R^*_{\mathscr{H}}$. Figure 12 allows us to investigate how $R^*_{\mathscr{H}}$ is affected by different noise levels (ochre line). Let d_c be the depth at

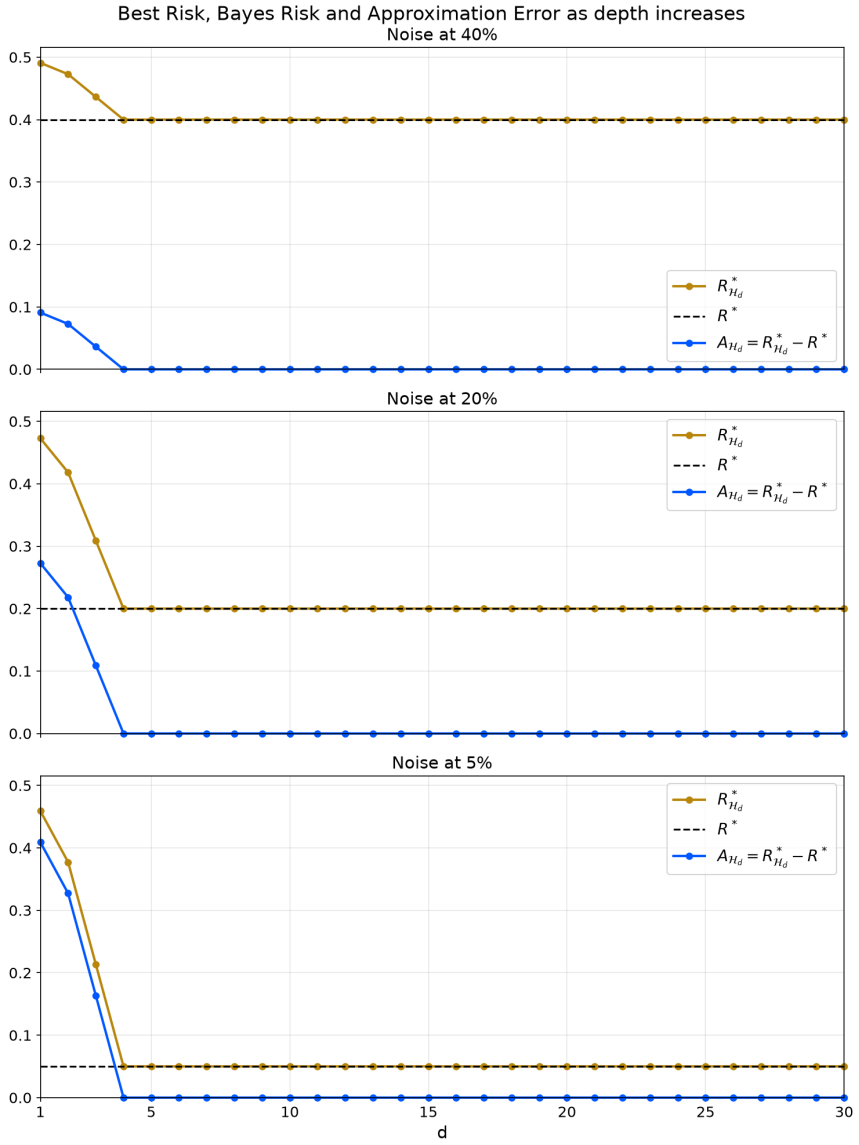


FIGURE 12

which the best risk saturates, which is the same in the three graphs. We make two observations:

(1) for $d < d_c$, the best risk decreases faster with less noise;

(2) for $d \geq d_c$, the best risk saturates and the saturating level is lower with less noise.

The first observation follows from the label-flip model: correcting a region reduces the risk by its probability mass times $2p-1$. Indeed, predicting the noiseless label errs with probability $1-p$, whereas predicting the opposite one errs with probability p, and the reduction in risk is $p-(1-p)=2p-1$. This factor shrinks as the flip probability $1-p$ grows, hence increasing the depth reduces the best risk more when there is less noise. The second observation is directly determined by the irreducible noise in the data. In fact, the best risk saturates exactly at the Bayes risk (black, dashed line).

Altogether, these two mechanisms explain the sensitivity of the Noise bound $\mathcal{B}_{Noise}$ to different noise levels. The second one, in particular, explains why the curve shifts upwards when there is more noise, similarly to the generalization gap. However, because also the cardinality of the hypothesis class and some constants depending on $n = 500$ enter the Noise bound, the change in resulting curve is smaller than the change in the best risk.

6. Conclusions

This thesis told a story of theory and evidence: we started from a bound using Hoeffding's inequality and a union bound over the hypothesis class, which did not capture the saturation of the empirical error curve for binary decision trees. Then, we took into account that measuring the complexity of the class of functions that decision trees can learn after they have seen the data is more representative of how the algorithm works in practice. By including this in our analysis via Lemma 4.10, we obtained a saturating bound. Finally, we noticed that the generalization performance of decision trees is affected by the level of noise in the data. Thus, using more refined concentration inequalities, we obtained a bound that is a function of the lowest risk that a decision tree can attain, which depends on the noise through the Bayes risk. This, combined with the result on the actual complexity of the hypothesis class, gave a bound that saturates and that is sensitive to the noise.

The main limitation of this thesis is the lack of an analysis of the difference between the selected model that we consider for theoretical results, i.e. the Empirical Risk Minimizer, and the model learned in practice by the DecisionTreeClassifier algorithm. In fact, this is a greedy algorithm that does not always select the best possible model. Among the results we derived, only the lower bound in Theorem 3.1 and the results from Theorem 5.2 onwards are specific to the ERM. Analyzing this issue in more depth might be a direction for future work.

Besides this, although we obtain a bound that saturates and is noise-sensitive, there is clearly room for improving this study and obtaining functions that capture the qualitative behavior of the generalization gap also under other aspects. In this regard, we believe that a promising direction is to study the qualitative descriptive power of bounds based on a *localized* version of the Rademacher complexity, proposed by [3]. Another potential development is to consider other measures of complexity of the hypothesis class, not necessarily based on the maximum depth.

Finally, we hope that the framework and approach proposed in this work might be applied to other algorithms in the future. Specifically, a natural next step would be Random Fourier Features, while we propose neural networks trained with Stochastic Gradient Descent as a more ambitious one.

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Appendix A. Explorative Experiments

A.1. **Noise Structure.** We generate the labels with a centered parabola (see Section A.2), so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Then, we inspect what happens if we use a Bernoullian noise instead of a Gaussian one, i.e. when data is generated as

$$y \leftarrow sign(f^*(x))b$$

for $b \sim \text{Bernoulli}(p)$. Notice that this noise flips the labels with a probability of $1 - p$. While from the first experiments it seemed that the algorithm achieved a higher test error, we postulate that this may be due to the portion of misclassified inputs, rather than to the noise structure itself. Hence, fixing $p = 0.8$, we look for the value of σ that achieves roughly the same portion of misclassified inputs (20%). Figure 13 shows that this happens for $\sigma = 6$, and Figures 14 and 15 show that the two noise structures then yield essentially the same train and test errors.

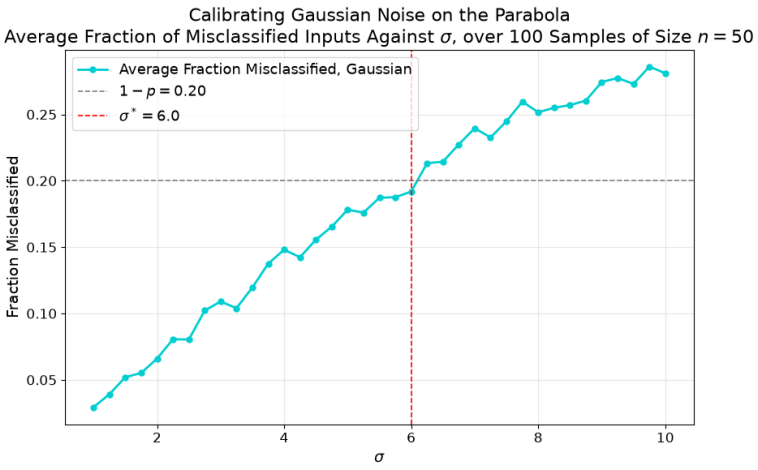


Figure 13

A.2. **Data-Generating Function.** Our first choice of f^* is a parabola centered horizontally at 5, the midpoint of the support $\mathcal{X}$, and vertically so that roughly half of the points fall above the x-axis and half below. I.e.,

$$f^*(x) = (x - 5)^2 - 6.25$$

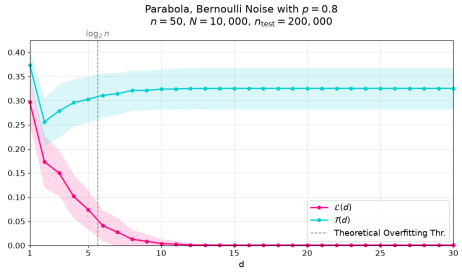


FIGURE 14

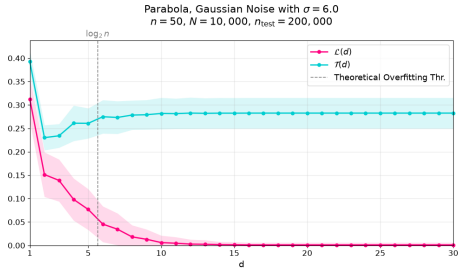


FIGURE 15

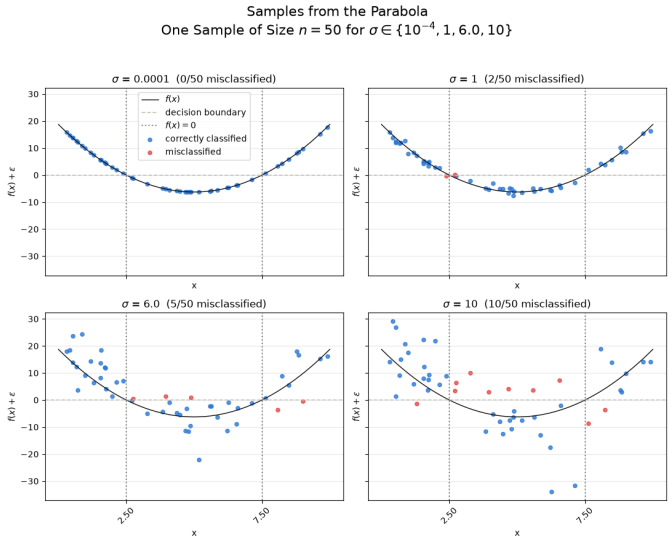


FIGURE 16

Figure 16 shows the data generated with the centered parabola, for different amounts of noise.

Because in theory an algorithm could learn such a classification pattern as soon as it sees 3 points (one point for each region), we investigate what happens when the underlying function is more complex. We study a sinusoid with varying frequency,

$$f^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

We multiply the sine by 5 to allow for enough height to properly inspect the data, while the argument of the sine is such that m is the number of times f^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$.

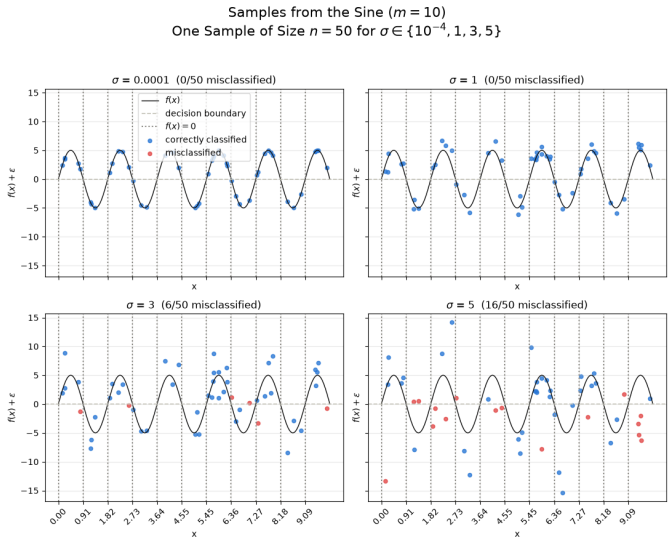


FIGURE 17

Figure 17 shows the data generated with the sine for $m = 10$, for different amounts of noise.

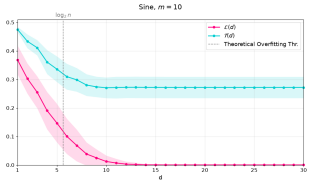


FIGURE 18

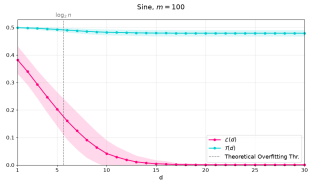


FIGURE 19

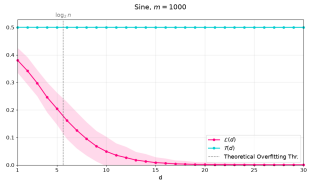


FIGURE 20

We explore how a random sample and the performance of the algorithm change for different sine frequencies. In particular, we expect that for a high frequency the algorithm will not learn, as our datasize is $n = 50$. In Figures 18–20, we test a function that oscillates $10, 100$, and 1000 times with a noise level of $\sigma = 3$ (we reduce it compared to $\sigma = 5$, having increased the complexity of the function). We notice that for $m = 1000$ the algorithm fails to learn, and the test error stays stable at 0.5. This is because the labels change so frequently with respect to the number of points in the training dataset n that it is as if we had random labels. Hence, we investigate the relation between the complexity of the underlying function m and

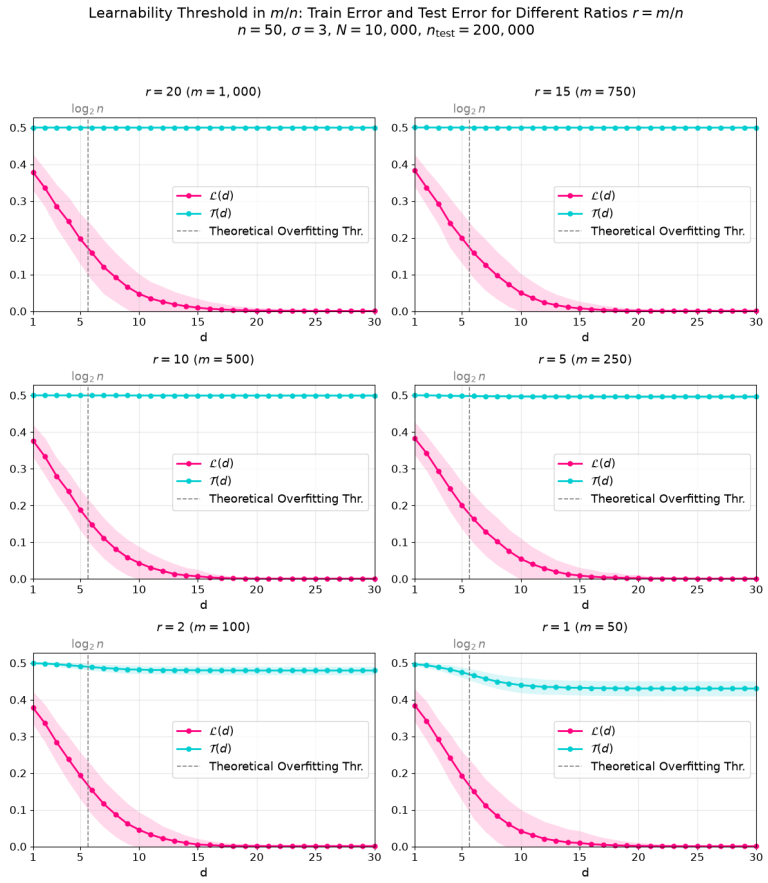


FIGURE 21

the data size n. We find that for $\frac{m}{n} > 5$, roughly, the algorithm ceases to learn (Figure 21). In the following experiments, we fix $m = 10$ and $n = 50$, which is very far from the threshold.

We then verify whether a Gaussian noise with $\sigma = 6$ achieves the same portion of misclassified inputs on average as a Bernoullian noise with $p = 0.8$, as was the case for data generated with a parabola. Figure 22 shows that this is not the case on average (31% against 20%), while Figure 23 shows that the desired value is $\sigma = 3.5$.

Finally, we investigate the aforementioned intuition about how easy it is for the algorithm to learn a simple function. We take $m = 2$ (which corresponds to a parabola, in terms of classification pattern), sample uniformly from $\mathcal{X}$, and look

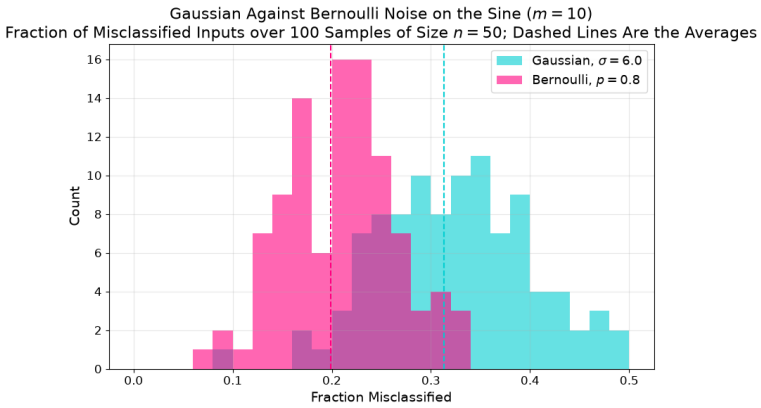


FIGURE 22

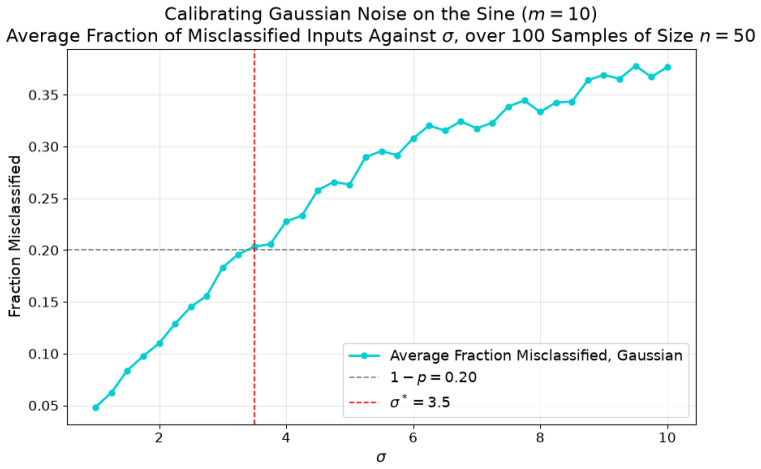


FIGURE 23

for the training set size n for which the test error gets within 0.01 of the Bayes risk on average, at depth 2 and 3.

We start with little noise, $\sigma = 1$, for which the Bayes risk is ≈ 0.05 (Figure 24). Here the algorithm learns the function quickly: with $n = 50$ points the test error is already ≈ 0.07, within 0.02 of the Bayes risk, and it gets within 0.01 with $n = 500$ at both depth 2 and 3.

With our noise level, $\sigma = 3.5$, the Bayes risk is ≈ 0.20 and $n = 1000$ training points at depth 2 and $n = 2000$ at depth 3 are needed on average for the test error to get

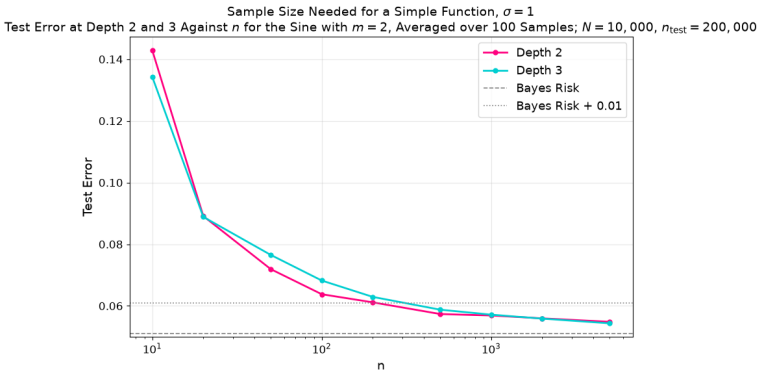


FIGURE 24

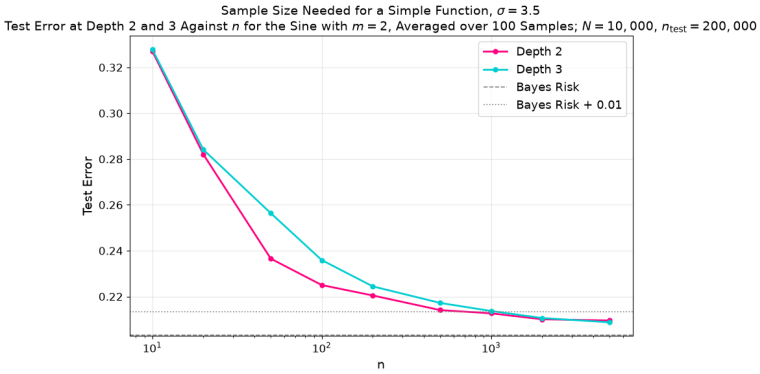


FIGURE 25

within 0.01 of it (Figure 25). Hence, although two splits are enough to represent the function, with this amount of noise the algorithm needs many points to place them correctly.

Appendix B. Details on the Experimental Design

B.1. **Hyperparameters.** Table 1 lists the hyperparameters used in all the experiments, unless stated otherwise. The noise level σ is chosen so that the average flip probability $\frac{1}{N}\sum_j \Phi\left(-|f^*(\bar{x}_j)|/\sigma\right)$ equals the target percentage of flipped labels, where Φ is the cumulative distribution function of the standard Gaussian. See Appendix B.2 for the computation of the constants.

Parameter	Symbol	Value
Support size	N	10^9
Train size	n	500
Test size	n_{test}	$200,000$
Iterations	T	100
Crossings of f^*	m	10
Noise	σ	≈ 3.5, 20% flipped labels
Depths	d	$1, \ldots, 30$
Constant, union bound	C_{union}	$6 \cdot 10^{-4}$
Constant, Rademacher–Massart	C_{radmass}	0.125
Constant, noise bound	C_{noise}	0.01
Seed		4

Table 1. Hyperparameters of the experiments.

B.2. **Fitting Constants.**

Union bound. We just rescale the bound so that it lies in the interval $[0, 1]$. For d large, $\log|\mathscr{H}_d| = N \log 2$. This means that the bound is $2\sqrt{\frac{N \log 2}{2n}}$, hence we need

$$C_{\text{union}} \cdot 2\sqrt{\frac{N \log 2}{2n}} = 1$$

Therefore we set

$$C_{\text{union}} = \frac{1}{2\sqrt{\frac{10^9 \log 2}{2 \cdot 500}}} = \frac{1}{2\sqrt{10^6 \log 2}} = \frac{1}{2\sqrt{\log 2}} 10^{-3} \approx 0.0006$$

Rademacher and Massart bound. Using the code `source/experiments.py`, we print the value of the generalization gap for the highest depth $d = 30$, getting

$$0.2860 \pm 0.0121$$

Therefore, to obtain that the curve of the Rademacher bound saturates at the same level, we need

$$C_{\text{RadMass}} 2\sqrt{2 \log 2} = 0.286 \quad \Rightarrow \quad C_{\text{RadMass}} = \frac{0.286}{2\sqrt{2 \log 2}} = 0.12145302 \approx 0.125$$

which we round up for the bound to lie above the curve.

Noise bound. Similarly, we want the bound to saturate at the same level of the empirical gap curve. The saturating level of the bound, i.e. its value for a large depth, is

$$4\sqrt{R^*_{\mathscr{H}_d} M} + 20M \qquad \text{with } M = \frac{(2n+2)\log 2 + 1}{n}$$

hence we set

$$C_{\text{noise}}(4\sqrt{R^*_{\mathscr{H}_d} M} + 20M) = 0.286.$$

We have

$$M = \frac{(1002)\log 2 + 1}{500} \approx 1.3910$$

hence

$$(4\sqrt{R^*_{\mathscr{H}_d} M} + 20M) \approx 4\sqrt{0.2 \cdot 1.391} + 20 \cdot 1.391 \approx 29.9298$$

Thus,

$$C_{\text{noise}} = 0.286/29.9298 \approx 0.01$$

where we use $R^*_d = 0.2$ the Bayes risk, that is the value attained by the best risk for a large enough depth.

B.3. **Computation of the Best Risk.** The algorithm is explained using one-based indexing for the sake of exposition, while the real code uses zero-based indexing as usual in Python. We produce N noiseless labels for each input value in the support and we cluster the intervals over $\mathcal{X}$ where inputs have the same label. Let L be the number of contiguous blocks we obtain. We create a `dp` matrix with dimension $k_{\max} \times 2$: in the columns, we have the labels $y_j \in \{1, -1\}$ for $j = 1, 2$; in the rows, we have the number of leaves in the tree $k = 1, \ldots, k_{\max}$, with $k_{\max} = \min(2^{d_{\max}}, L)$ the maximum number of leaves and $d_{\max}$ the largest depth we consider overall (that is $d_{\max} = 30$, see Section 3). Note that the tree

reproducing exactly the noiseless labels has L leaves and has expected risk equal to $1-p$, the portion of corrupted labels, reaching the Bayes risk. Since the best risk is bounded from below by the Bayes risk, considering more than L leaves cannot improve it. So, to avoid useless computations, we consider the minimum between L and $2^{d_{\max}}$. Given the true labels for each of the L blocks, it is easy to create also a `cost` matrix with dimension $L \times 2$, where `cost[i, j]` is the risk given by predicting y_j at block i. Let y_i be the true label in block i and ρ_i the portion of inputs falling into block i. We have

$$\texttt{cost[i,j]} = \begin{cases} \rho_i \cdot (1-p) & \text{if } y_i = y_j \\ \rho_i \cdot p & \text{if } y_i \neq y_j \end{cases}$$

We initialize the `dp` matrix setting all entries to ∞. For the first iteration over block $i = 1$, we set

$$\texttt{dp[1, :]} = \texttt{cost[1, :]}$$

For each iteration $i = 2, \dots, L$ we dynamically update the `dp` matrix with minimum risks. Dynamically updating means that in `dp[k,j]` we store the cumulative minimum risk of having predicted label y_j at the *previous* block $i-1$ with k leaves, and we create a new $k_{\max} \times 2$ matrix called `new` such that in `new[k,j]` we store the minimum risk of predicting label y_j at the current block i with k leaves. This minimum risk is computed as follows. Given that we predict y_j at block i, we choose between the tree that has predicted label y_j at block $i-1$ using k leaves, i.e. that keeps the same label, and the tree that has predicted label $-y_j$ at block $i-1$ using $k-1$ leaves, i.e. that switches labels. These two values are stored in the `dp` matrix, as said. Then, we add the cost of predicting label y_j in the current block i. This is summarized by the following line

$$\texttt{new[k,j]} = \min(\texttt{dp[k,j]}, \texttt{dp[k-1,3-j]}) + \texttt{cost[i,j]}$$

Note that $3-j$ swaps the indices 1 and 2, i.e. it stands for $-y_j$. Also, this code holds for $k > 1$, while for $k = 1$ the cost of switching is set to ∞, as it is not admissible. After passing over all $2k_{\max}$ entries of `new`, we update `dp = new` and pass to the next block $i+1$. The `dp` matrix will now contain the cumulative minimum risks for each possible label predicted in the last block i. Note that using an auxiliary matrix

`new` is not strictly necessary, as we could update the `dp` matrix recursively, but using it eases understanding the logic. In the end, after passing over the last contiguous block, we obtain a matrix with the minimum risk for each number of leaves and each label assigned to the last block. We create a $k_{\max}$-long list `exact_k_risk` with the minimum value for each k,

$$\texttt{exact_k_risk[k]} = \min_{j \in \{1,2\}} \texttt{dp[k,j]}$$

Then, we create another $k_{\max}$-long list `at_most_k_risk` that stores the cumulative minimum for each possible *maximum* number of leaves,

$$\texttt{at_most_k_risk[k]} = \min_{r \in [1,k]} \texttt{exact_k_risk[r]}$$

Thus, the k-th entry of `at_most_k_risk` contains the best risk allowing at most k leaves. Finally, for each depth d we set $k_d = \min(2^d, L)$ and select the k_d-th entry in `at_most_k_risk`, yielding the best risk among trees with maximum depth d.